OGC® DOCUMENT: 26-042

External identifier of this OGC® document: http://www.opengis.net/doc/IS/indoorgml-xml/2.0

Open Geospatial Consortium

OGC INDOORGML 2.0 PART 2A – XML ENCODING

STANDARD
Implementation

CANDIDATE SWG DRAFT

Version: 0.5.0
Submission Date: 2025-08-06
Approval Date: 2099-01-01
Publication Date: 2099-01-01
Editor: Abdoulaye Diakité, Taehoon Kim, Sisi Zlatanova, Ki-Joune Li, Zhiyong Wang

Notice for Drafts: This document is not an OGC Standard. This document is distributed for review and comment. This document is subject to change without notice and may not be referred to as an OGC Standard.
Recipients of this document are invited to submit, with their comments, notification of any relevant patent rights of which they are aware and to provide supporting documentation.

Open Geospatial Consortium

License Agreement

Use of this document is subject to the license agreement at https://www.ogc.org/license

Suggested additions, changes and comments on this document are welcome and encouraged. Such suggestions may be submitted using the online change request form on OGC web site: http://ogc.standardstracker.org/

Copyright notice

Copyright © 2026 Open Geospatial Consortium
To obtain additional rights of use, visit https://www.ogc.org/legal

Note

Attention is drawn to the possibility that some of the elements of this document may be the subject of patent rights. The Open Geospatial Consortium shall not be held responsible for identifying any or all such patent rights.

Recipients of this document are requested to submit, with their comments, notification of any relevant patent claims or other intellectual property rights of which they may be aware that might be infringed by any implementation of the standard set forth in this document, and to provide supporting documentation.

CONTENTS

I. ABSTRACT v

II. KEYWORDS v

III. PREFACE vi

IV. SECURITY CONSIDERATIONS vii

V. SUBMITTING ORGANIZATIONS viii

VI. SUBMITTERS viii

1. SCOPE 2

2. CONFORMANCE 4

3. NORMATIVE REFERENCES 6

4. TERMS AND DEFINITIONS 8

5. CONVENTIONS 11
   - 5.1. Identifiers 11
   - 5.2. Use of XML terminology 11

6. OVERVIEW OF THE XML ENCODING 13
   - 6.1. Design Principle 13

7. INDOORXML CORE 16
   - 7.1. General requirements for encoding standard 16
   - 7.2. IndoorFeatures 17
   - 7.3. ThematicLayer 18
   - 7.4. PrimalSpaceLayer 20
   - 7.5. CellSpace 22
   - 7.6. CellBoundary 24
   - 7.7. DualSpaceLayer 26
   - 7.8. Node 28
   - 7.9. Edge 29
   - 7.10. InterLayerConnection 30
   - 7.11. ExternalReference 33

8. INDOORXML NAVIGATION 36

8.1. General requirements for encoding standards 36
8.2. NavigableSpace 37
8.3. GeneralSpace 38
8.4. TransferSpace 40
8.5. NavigableBoundary 42
8.6. NonNavigableSpace 43
8.7. ObjectSpace 44
8.8. NonNavigableBoundary 46
8.9. Route 47

ANNEX A (NORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE 50
A.1. Conformance Class Core 50
A.2. Conformance Class Navigation Extension 54

ANNEX B (NORMATIVE) XML SCHEMA FOR INDOORGML 2 60
B.1. IndoorGML Core Module 60
B.2. IndoorGML Indoor Navigation Module 68

ANNEX C (INFORMATIVE) EXAMPLE OF CODELIST 74
C.1. GeneralSpace 74
C.2. TransferSpace 78

BIBLIOGRAPHY 81

I ABSTRACT

The IndoorGML 2.0 Part 2a – XML Encoding Standard presents the implementation-dependent Geography Markup Language (GML) XML schema encoding of the concepts defined by the IndoorGML 2.0 Part 1 – Conceptual Model Standard (OGC 22-045r5). The XML Encoding is compliant with GML version 3.2.1 as specified by OGC 07-036r1. The XML encoding of IndoorGML 2.0 defines a concrete implementation schema for representing indoor spatial information, including primal space, dual space, thematic layers, interlayer connections, and navigation-related indoor features. This encoding can be used to store and exchange 2D and/or 3D indoor space and indoor navigation network models in XML format as data sets or via web services.

This Part 2a of the Standard defines how the concepts developed in Part 1 are realized using XML Schema definitions. The XML encoding represents a full mapping of the Conceptual Model and is derived from the UML model following the UML-to-GML/XML encoding rules. Table 1 maps requirements classes from the IndoorGML 2.0 Conceptual Model into the implementation details documented in this standard.

Table 1 — Requirements classes mapping

CONCEPTUAL MODEL	SECTION	XML SCHEMA
IndoorGML Core	Clause 7	Annex B.1
Navigation Extension	Clause 8	Annex B.2

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

OGCDoc, OGC documents, indoor, navigation, IndoorGML, GML, XML

PREFACE

In order to achieve consensus on the modeling indoor spaces and their neighborhood relationships to support indoor location-based services, a UML Conceptual Model, IndoorGML 2.0 Part 1, was approved as an OGC Standard (OGC 22-045r5) in June, 2025. This model covers geometric and semantic properties of indoor spaces relevant for indoor navigation. This IndoorGML 2.0 Part 2a: XML Encoding Standard defines how those concepts should be realized using XML schema format. The XML encoding is intended for software developers, data providers, standards implementers, and service providers who need a structured, standard, and machine-readable representation of IndoorGML datasets using XML/GML.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this Standard.

V SUBMITTING ORGANIZATIONS

The following organizations submitted this Document to the Open Geospatial Consortium (OGC):

- The University of New South Wales
- Pusan National University
- CityGeometrix
- National Institute of Advanced Industrial Science and Technology
- SpatialAlgebra

VI SUBMITTERS

All questions regarding this submission should be directed to the editor or the submitters:

NAME	AFFILIATION	EMAIL	OGC MEMBER
Abdoulaye Diakité	CityGeometrix Pty Ltd	abdou@citygeometrix.com	Yes
Taehoon Kim	National Institute of Advanced Industrial Science and Technology	kim.taehoon@aist.go.jp	Yes
Sisi Zlatanova	University of New South Wales	s.zlatanova@unsw.edu.au	Yes
Ki-Joune Li	Pusan National University	lik@pnu.edu	Yes
Zhiyong Wang	SpatialAlgebra Technology Limited	zhiyongwang@spatialalgebra.tech	Yes

1 SCOPE

1 SCOPE

The OGC® IndoorGML 2.0 Part 2a – XML Encoding Standard defines an XML encoding for IndoorGML 2.0 Part 1 — Conceptual Model. The encoding is implemented as an application schema of the Geography Markup Language version 3.2.1 (GML 3.2.1).

The IndoorGML XML Encoding Standard specifies:

- XML schema for IndoorGML 2.0 core module;
- XML schema for IndoorGML 2.0 navigation module;
- the geometry encoding rules based on GML 3.2.1 geometry objects; and
- a statement to which IndoorGML XML conformance classes an IndoorGML 2.0 instance conforms to.

IndoorGML XML encoding is an implementation encoding of IndoorGML 2.0 Part 1. Therefore, IndoorGML XML instances shall be interpreted according to the semantics defined in the IndoorGML 2.0 conceptual model.

2 CONFORMANCE

2 CONFORMANCE

This standard defines implementation specification which specifies how the IndoorGML 2.0 Conceptual Model should be implemented using Extensible Markup Language (XML) and GML 3.2.1. The Standardization Target for this standard is:

- Implementations of the IndoorGML 2.0 Conceptual Model using XML encodings.

Conformance with this standard shall be checked using all the relevant tests specified in Annex A (normative) of this document. The framework, concepts, and methodology for testing, and the criteria to be achieved to claim conformance are specified in the OGC Compliance Testing Policies and Procedures and the OGC Compliance Testing web site.

In order to conform to this OGC® interface standard, a software implementation shall choose to implement:

- Any one of the conformance levels specified in Annex A (normative).

All requirements-classes and conformance-classes described in this document are owned by the standard(s) identified.

Table 2 — Conformance classes

CONFORMANCE CLASS	URI
IndoorGML Core	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core
Navigation Extension	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi

3 NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

Data elements and interchange formats – Information interchange – Representation of dates and times

Geographic information – Reference model – Part 1: Fundamentals

Geographic Information – Conceptual Schema Language

Geographic information – Conformance and testing

Geographic Information – Spatial Schema

Geographic Information – Rules for Application Schemas

Geographic information – Spatial referencing by coordinates

Geographic Information – Metadata – Part 1: Fundamentals

Geographic Information – Metadata – XML schema implementation

George Percivall, Open Geospatial Consortium (OGC): OGC 04-084r4, *Topic 0 — Overview*. Open Geospatial Consortium (2020). http://www.opengis.net/doc/AS/topic-0/9.0.

Cliff Kottman, Carl Reed, Open Geospatial Consortium (OGC): OGC 08-126, *Topic 5 — Features*. Open Geospatial Consortium (2009). https://docs.ogc.org/as/08-126/08-126/pdf.

Cliff Kottman, Open Geospatial Consortium (OGC): OGC 99-108r2, *Topic 8 — Relationships Between Features*. Open Geospatial Consortium (1999). https://docs.ogc.org/as/99-108r2/99-108r2/pdf.

Cliff Kottman, Open Geospatial Consortium (OGC): OGC 99-110, *Topic 10 — Feature Collections*. Open Geospatial Consortium (1999). https://docs.ogc.org/as/99-110/99-110/pdf.

Clemens Portele, Open Geospatial Consortium (OGC): OGC 07-036r1, *OpenGIS Geography Markup Language (GML) Encoding Standard — with corrigendum*. Open Geospatial Consortium (2018). https://docs.ogc.org/is/07-036r1/07-036r1/pdf.

Sisi Zlatanova, Abdoulaye Diakite, Taehoon Kim, Ki-Joune Li, Open Geospatial Consortium (OGC): OGC 22-045r5, *OGC IndoorGML 2.0 Part 1 – Conceptual Model*. Open Geospatial Consortium (2025). http://www.opengis.net/doc/IS/indoorgml/2.0.

4 TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

This document uses the terms defined in OGC Policy Directive 49, which is based on the ISO/IEC Directives, Part 2, Rules for the structure and drafting of International Standards. In particular, the word “shall” (not “must”) is the verb form used to indicate a requirement to be strictly followed to conform to this document and OGC documents do not use the equivalent phrases in the ISO/IEC Directives, Part 2.

This document also uses terms defined in the OGC Standard for Modular specifications (OGC 08-131r3), also known as the ‘ModSpec’. The definitions of terms such as standard, specification, requirement, and conformance test are provided in the ModSpec.

For the purposes of this document, the following additional terms and definitions apply.

4.1. conceptual model

model that defines concepts of a universe of discourse

[**SOURCE:** ISO 19101-1:2014]

4.2. conceptual schema

formal description of a conceptual model

[**SOURCE:** ISO 19101-1:2014]

4.3. feature

abstraction of real world phenomena

Note 1 to entry: Features are objects that have an identity that allows for distinguishing features from each other. Features can have spatial and non-spatial properties that describe the features in more detail. Spatial properties are properties that have a geometry from ISO 19107:2003 as data type.

Note 2 to entry: Features are denoted by the stereotype «FeatureType» in the IndoorGML 2.0 Conceptual Model.

[**SOURCE:** ISO 19101-1:2014]

5 CONVENTIONS

5 CONVENTIONS

5.1. Identifiers

The normative provisions in this standard are denoted by the URI

http://www.opengis.net/spec/indoorgml-2/2.0/xml

All requirements and conformance tests that appear in this document are denoted by partial URIs which are relative to this base.

5.2. Use of XML terminology

The following terms and concepts are used as specified in the W3C XML specifications and OGC GML 3.2.1:

- “element” refers to an XML element tag;
- “attribute” refers to a name/value attribute on an XML element;
- “complexType” refers to an XML Schema complex type definition;
- “substitutionGroup” refers to the GML element substitution mechanism allowing specialized elements to substitute base GML feature types; and
- “property element” refers to an XML element that represents an association between features or objects.

6

OVERVIEW OF THE XML ENCODING

6 OVERVIEW OF THE XML ENCODING

The XML encoding of IndoorGML 2.0 implements the conceptual classes of IndoorGML 2.0 Part 1 as GML-compliant XML elements and types. The conceptual model remains normative for semantics, while this document defines the XML schema representation.

6.1. Design Principle

The XML encoding is based on the following encoding principles.

6.1.1. Conceptual model preservation

The XML encoding of the IndoorGML core module preserves the IndoorGML 2.0 core conceptual model structure, as shown in Figure 1.

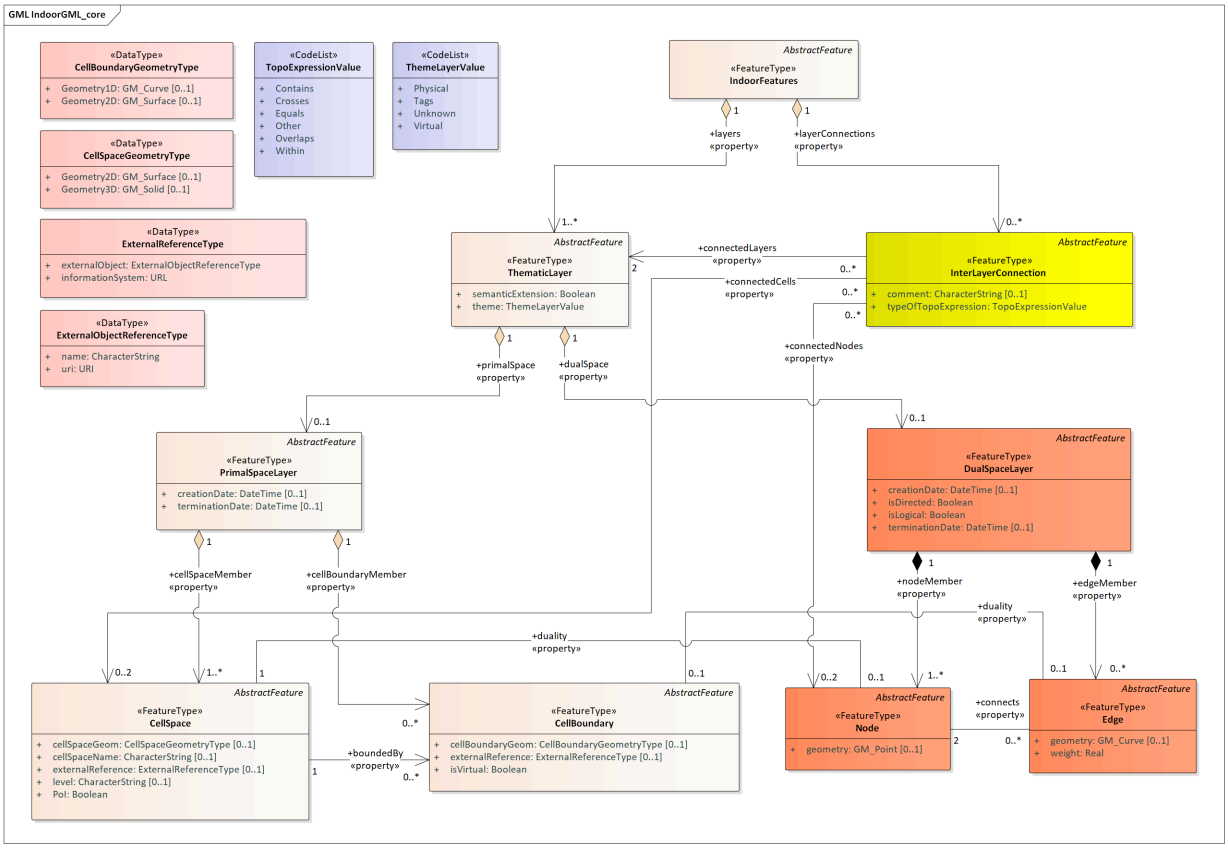


Figure 1 — IndoorGML 2.0 core module structure

Each feature class is mapped to an XML element inheriting from or substituting GML feature elements (e.g. gml:AbstractFeature). For example, the <CellSpace> element substitutes gml:AbstractFeature and extends gml:AbstractFeatureType to represent cell spaces.

The XML encoding of the IndoorGML navigation module preserves the IndoorGML 2.0 navigation conceptual model structure, as shown in Figure 2.

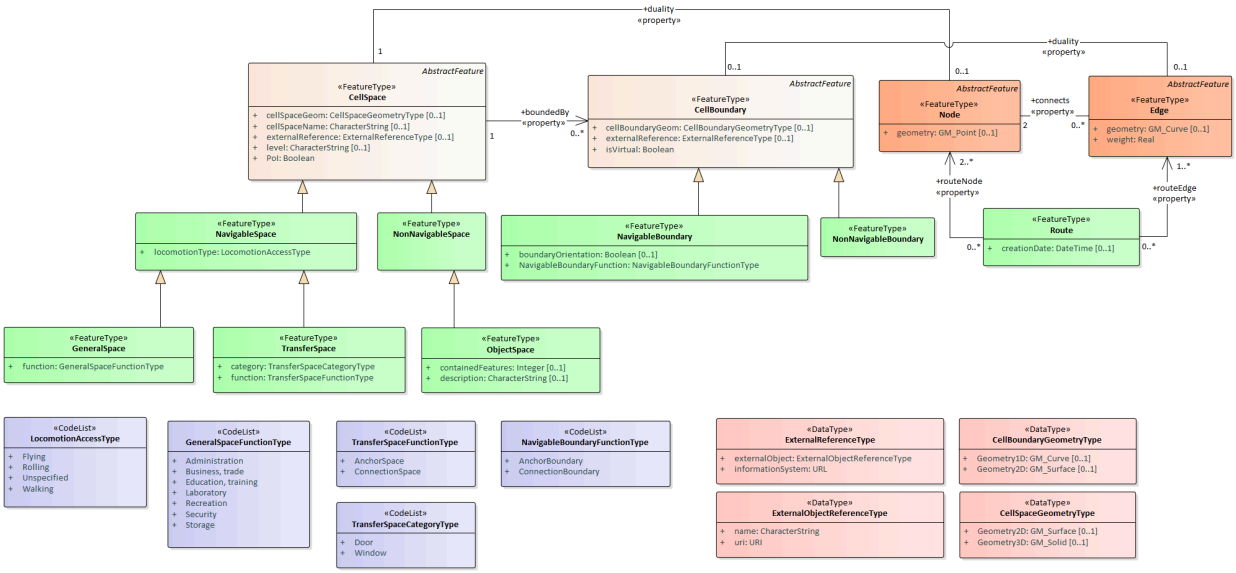


Figure 2 — IndoorGML 2.0 navigation module structure

Navigation feature classes extend core elements using GML inheritance patterns, implementing substitutions where appropriate. For example, <NavigableSpace> substitutes <CellSpace> and extends its complex type definition.

6.1.2. XML Schema design workflow

The XML encoding schema is derived from the IndoorGML 2.0 UML model and is designed on the basis of OGC GML 3.2.1. XML schema (.xsd) documents are generated from the UML model, establishing elements and complex types for the Core and Navigation modules. Typical patterns include converting associations into GML property types using xlink:href references (for example, the duality link from CellSpace to Node), mapping attributes to native XML schema elements (such as creationDate of type xs:dateTime), and representing UML code lists as XML simple types restricting xs:string with enumeration constraints.

7

INDOORXML CORE

7 INDOORXML CORE

REQUIREMENTS CLASS 1	
IDENTIFIER	http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
TARGET TYPE	Implementation Specification
CONFORMANCE CLASS	Conformance class A.1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core
PREREQUISITES	http://www.opengis.net/spec/indoorgml/2.0/req http://www.opengis.net/spec/gml/3.2.1/req
NORMATIVE STATEMENTS	Requirement 1: /req/core/global Requirement 2: /req/core/indoorfeatures Requirement 3: /req/core/thematiclayer Requirement 4: /req/core/primalspacelayer Requirement 5: /req/core/cellspace Requirement 6: /req/core/cellboundary Requirement 7: /req/core/dualspacelayer Requirement 8: /req/core/node Requirement 9: /req/core/edge Requirement 10: /req/core/interlayerconnection

7.1. General requirements for encoding standard

The IndoorGML Core module defines the foundational features and data structures for representing indoor spaces. It is composed of three main parts:

- The primal space, describing cellular space subdivisions and their boundaries;
- The dual space, carrying network graph information (nodes and edges); and
- The inter-layer connections, describing topological relationships between thematic layers.

An instance XML document representing an IndoorGML 2.0 Core dataset SHALL be well-formed XML and conform to the XML schemas defined in this standard and OGC GML 3.2.1.

REQUIREMENT 1: REQUIREMENT FOR GLOBAL XML SCHEMA VALIDATION	
IDENTIFIER	/req/core/global
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	An instance XML document representing an IndoorGML 2.0 Core dataset SHALL be well-formed XML and validate against the schemas defined in this standard (including IndoorGML_core.xsd and OGC GML 3.2.1).

7.2. IndoorFeatures

The IndoorFeatures element is an instance of IndoorFeatures, representing the root container feature for an indoor dataset. It aggregates one or more thematic layers and may optionally include interlayer connections.

The IndoorFeatures element shall contain:

- gml:id attribute
- layers property containing or referencing ThematicLayer elements
- optionally, layerConnections property containing or referencing InterLayerConnection elements

```
<element name="IndoorFeatures" substitutionGroup="gml:AbstractFeature" type="core:
IndoorFeaturesType"/>
<complexType name="IndoorFeaturesType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element maxOccurs="unbounded" name="layers">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:ThematicLayer"/>
        </sequence>
        <attributeGroup ref="gml:AssociationAttributeGroup"/>
       </extension>
      </complexContent>
     </complexType>
    </element>
    <element maxOccurs="unbounded" minOccurs="0" name="layerConnections">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:InterLayerConnection"/>
        </sequence>
```

```
            <attributeGroup ref="gml:AssociationAttributeGroup"/>
           </extension>
          </complexContent>
         </complexType>
        </element>
       </sequence>
      </extension>
     </complexContent>
    </complexType>
    <complexType name="IndoorFeaturesPropertyType">
     <sequence minOccurs="0">
      <element ref="core:IndoorFeatures"/>
     </sequence>
     <attributeGroup ref="gml:AssociationAttributeGroup"/>
     <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
```

Listing — XML schema for the IndoorFeatures element

```
<core:IndoorFeatures gml:id="IF_1"
   xmlns:core="http://www.opengis.net/indoorgml/2.0/core"
   xmlns:gml="http://www.opengis.net/gml/3.2"
   xmlns:xlink="http://www.w3.org/1999/xlink">
   <!-- mandatory properties: at least one ThematicLayer -->
   <core:layers>
      <core:ThematicLayer gml:id="TL_Pedestrian">...</core:ThematicLayer>
   </core:layers>
   <core:layers>
      <core:ThematicLayer gml:id="TL_Wheelchair">...</core:ThematicLayer>
   </core:layers>

   <!-- optional properties: InterLayerConnection -->
   <core:layerConnections>
      <core:InterLayerConnection gml:id="ILC_1">...</core:InterLayerConnection>
   </core:layerConnections>
</core:IndoorFeatures>
```

Listing — An example of the IndoorFeatures element

REQUIREMENT 2: REQUIREMENT FOR INDOORFEATURES XML ELEMENT	
IDENTIFIER	/req/core/indoorfeatures
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	An IndoorFeatures element SHALL contain at least one layers property containing or referencing a ThematicLayer element.
B	When present, each layerConnections property of an IndoorFeatures element SHALL contain or reference an InterLayerConnection element.
C	An IndoorFeatures element SHALL contain a unique gml:id attribute.

7.3. ThematicLayer

The ThematicLayer element is an instance of ThematicLayer, which represents one contextual decomposition of indoor space (e.g. physical layer, navigation layer, sensor layer).

The ThematicLayer element shall contain:

- gml:id attribute
- semanticExtension element indicating whether there is an extension module with additional semantic information
- theme element determining what type of model representation can be expected
- optionally, primalSpace property for the PrimalSpaceLayer element
- optionally, dualSpace property for the DualSpaceLayer element

```
<element name="ThematicLayer" substitutionGroup="gml:AbstractFeature" type="core:
ThematicLayerType"/>
<complexType name="ThematicLayerType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element name="semanticExtension" type="boolean"/>
    <element name="theme" type="core:ThemeLayerValueType"/>
    <element minOccurs="0" name="primalSpace">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:PrimalSpaceLayer"/>
        </sequence>
        <attributeGroup ref="gml:AssociationAttributeGroup"/>
       </extension>
      </complexContent>
     </complexType>
    </element>
    <element minOccurs="0" name="dualSpace">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:DualSpaceLayer"/>
        </sequence>
        <attributeGroup ref="gml:AssociationAttributeGroup"/>
       </extension>
      </complexContent>
     </complexType>
    </element>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="ThematicLayerPropertyType">
 <sequence minOccurs="0">
```

```
    <element ref="core:ThematicLayer"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the ThematicLayer element

```
<core:ThematicLayer gml:id="TL_1">
   <!-- mandatory properties -->
   <core:semanticExtension>true</core:semanticExtension>
   <core:theme>Physical</core:theme>

   <!-- optional properties -->
   <core:primalSpace xlink:href="#PSL_1"/>
   <core:dualSpace xlink:href="#DSL_1"/>
</core:ThematicLayer>
```

Listing — An example of the ThematicLayer element

REQUIREMENT 3: REQUIREMENT FOR THEMATICLAYER XML ELEMENT	
IDENTIFIER	/req/core/thematiclayer
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A ThematicLayer element SHALL declare whether there is an extension module with additional semantic information using the semanticExtension element (true or false).
B	A ThematicLayer element SHALL determine what type of model representation is expected using the theme element.
C	The theme value SHALL be one of the following: "Tags", "Virtual", "Unknown", "Physical", or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: WiFi" or "other: Sensor").
D	When present, the primalSpace property of a ThematicLayer element SHALL contain or reference (via xlink:href) a PrimalSpaceLayer element.
E	When present, the dualSpace property of a ThematicLayer element SHALL contain or reference (via xlink:href) a DualSpaceLayer element.
F	A ThematicLayer element SHALL contain a unique gml:id attribute.

7.4. PrimalSpaceLayer

The PrimalSpaceLayer element is an instance of PrimalSpaceLayer, which contains spatial objects representing indoor subdivisions in primal space.

The PrimalSpaceLayer element shall contain:

- gml:id attribute
- cellSpaceMember property for representing CellSpace elements
- optionally, cellBoundaryMember property for representing CellBoundary elements
- optionally, creationDate element
- optionally, terminationDate element

```
<element name="PrimalSpaceLayer" substitutionGroup="gml:AbstractFeature" type="core:
PrimalSpaceLayerType"/>
<complexType name="PrimalSpaceLayerType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="creationDate" type="dateTime"/>
    <element minOccurs="0" name="terminationDate" type="dateTime"/>
    <element maxOccurs="unbounded" name="cellSpaceMember">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:CellSpace"/>
        </sequence>
        <attributeGroup ref="gml:AssociationAttributeGroup"/>
       </extension>
      </complexContent>
     </complexType>
    </element>
    <element maxOccurs="unbounded" minOccurs="0" name="cellBoundaryMember">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:CellBoundary"/>
        </sequence>
        <attributeGroup ref="gml:AssociationAttributeGroup"/>
       </extension>
      </complexContent>
     </complexType>
    </element>
   </sequence>
  </extension>
 </complexContent>
</complexType>
```

Listing — XML schema for the PrimalSpaceLayer element

```
<core:PrimalSpaceLayer gml:id="PSL_1">
   <!-- mandatory properties (at least one CellSpace) -->
   <core:cellSpaceMember>
      <core:CellSpace gml:id="CS_1">...</core:CellSpace>
   </core:cellSpaceMember>

   <!-- optional properties -->
   <core:cellBoundaryMember>
      <core:CellBoundary gml:id="CB_1">...</core:CellBoundary>
   </core:cellBoundaryMember>
```

```
    <core:creationDate>2025-08-06T18:39:00Z</core:creationDate>
</core:PrimalSpaceLayer>
```

Listing — An example of the PrimalSpaceLayer element

REQUIREMENT 4: REQUIREMENT FOR PRIMALSPACELAYER XML ELEMENT	
IDENTIFIER	/req/core/primalspacelayer
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A PrimalSpaceLayer element SHALL contain at least one cellSpaceMember property containing or referencing a CellSpace element.
B	The CellSpace elements belonging to the same PrimalSpaceLayer SHALL not intersect with each other.
C	The CellBoundary elements belonging to the same PrimalSpaceLayer SHALL not intersect with each other.
D	When present, creationDate and terminationDate SHALL be formatted as valid xs: dateTime strings.
E	A PrimalSpaceLayer element SHALL contain a unique gml:id attribute.

7.5. CellSpace

The CellSpace element is an instance of CellSpace, which represents a unit of indoor space such as a room, corridor, stairwell, elevator cabin, or zone.

The CellSpace element shall contain:

- gml:id attribute
- PoI element specifying whether this CellSpace is a point of interest
- optionally, cellSpaceName element for the name assigned to the space
- optionally, level element
- optionally, duality property mapped to the corresponding Node element in the dual space
- optionally, cellSpaceGeom element specifying the 2D or 3D geometry of the cell space
- optionally, boundedBy property referencing the linked CellBoundary elements
- optionally, externalReference property for representing ExternalReference elements

```
<element name="CellSpace" substitutionGroup="gml:AbstractFeature" type="core:CellSpaceType"/>
<complexType name="CellSpaceType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="cellSpaceGeom" type="core:CellSpaceGeometryPropertyType"/>
    <element minOccurs="0" name="externalReference" type="core:ExternalReferencePropertyType"/>
    <element minOccurs="0" name="level" type="string"/>
    <element name="PoI" type="boolean"/>
    <element minOccurs="0" name="cellSpaceName" type="string"/>
    <element minOccurs="0" name="duality" type="core:NodePropertyType"/>
    <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type="core:
CellBoundaryPropertyType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="CellSpacePropertyType">
 <sequence minOccurs="0">
  <element ref="core:CellSpace"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type="core:
CellSpaceGeometryType"/>
<complexType name="CellSpaceGeometryType">
 <sequence>
  <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
  <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
 </sequence>
</complexType>
<complexType name="CellSpaceGeometryPropertyType">
 <sequence>
  <element ref="core:CellSpaceGeometry"/>
 </sequence>
</complexType>
```

Listing — XML schema for the CellSpace element

```
<core:CellSpace gml:id="CS_1">
   <!-- mandatory property -->
   <core:PoI>false</core:PoI>

   <!-- optional properties -->
   <core:cellSpaceGeom>
      <core:Geometry3D>
         <gml:Solid gml:id="CSG_CS_1">
            ...
         </gml:Solid>
      </core:Geometry3D>
   </core:cellSpaceGeom>
   <core:duality xlink:href="#ND_1"/>
   <core:level>3rd Floor</core:level>
   <core:cellSpaceName>Locker Room Lv3</core:cellSpaceName>
   <core:boundedBy xlink:href="#CB_1"/>
</core:CellSpace>
```

Listing — An example of the CellSpace element

REQUIREMENT 5: REQUIREMENT FOR CELLSPACE XML ELEMENT	
IDENTIFIER	/req/core/cellspace
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A CellSpace element SHALL declare whether it is a point of interest space using the PoI element (true or false).
B	When present, the duality property of a CellSpace element SHALL reference (via xlink:href) or contain the corresponding Node element in the dual space.
C	When present, each boundedBy property of a CellSpace element SHALL reference (via xlink:href) or contain a corresponding CellBoundary element.
D	The geometry of each linked CellBoundary element SHALL not exceed the extent of the CellSpace geometry (cellSpaceGeom).
E	A CellSpace element SHALL contain a unique gml:id attribute.

7.6. CellBoundary

The CellBoundary element is an instance of CellBoundary, representing a physical or virtual boundary of a cell space.

The CellBoundary element shall contain:

- gml:id attribute
- isVirtual element indicating whether a CellBoundary corresponds to a virtual surface (true) or a physical one (false)
- optionally, duality property mapped to the corresponding Edge element
- optionally, cellBoundaryGeom element specifying the 1D or 2D geometry of the cell boundary
- optionally, externalReference property for representing ExternalReference elements

```
<element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type="core:
CellBoundaryType"/>
<complexType name="CellBoundaryType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="cellBoundaryGeom" type="core:CellBoundaryGeometryPropertyTy
pe"/>
    <element minOccurs="0" name="externalReference" type="core:ExternalReferencePropertyType"/>
    <element name="isVirtual" type="boolean"/>
```

```
      <element minOccurs="0" name="duality" type="core:EdgePropertyType"/>
     </sequence>
    </extension>
  </complexContent>
</complexType>
<complexType name="CellBoundaryPropertyType">
  <sequence minOccurs="0">
    <element ref="core:CellBoundary"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject" type="core:
CellBoundaryGeometryType"/>
<complexType name="CellBoundaryGeometryType">
  <sequence>
    <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
    <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
  </sequence>
</complexType>
<complexType name="CellBoundaryGeometryPropertyType">
  <sequence>
    <element ref="core:CellBoundaryGeometry"/>
  </sequence>
</complexType>
```

Listing — XML schema for the CellBoundary element

```
<core:CellBoundary gml:id="CB_1">
    <!-- mandatory property -->
    <core:isVirtual>false</core:isVirtual>

    <!-- optional properties -->
    <core:cellBoundaryGeom>
       <core:Geometry2D>
          <gml:Surface gml:id="CBG_CB_1">
             ...
          </gml:Surface>
       </core:Geometry2D>
    </core:cellBoundaryGeom>
    <core:duality xlink:href="#EDG_1"/>
</core:CellBoundary>
```

Listing — An example of the CellBoundary element

REQUIREMENT 6: REQUIREMENT FOR CELLBOUNDARY XML ELEMENT	
IDENTIFIER	/req/core/cellboundary
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A CellBoundary element SHALL specify whether it represents a virtual boundary (true) or a physical boundary (false) using the isVirtual element.
B	When present, the duality property of a CellBoundary element SHALL reference (via xlink:href) or contain the corresponding Edge element in the dual space.

REQUIREMENT 6: REQUIREMENT FOR CELLBOUNDARY XML ELEMENT	
C	A CellBoundary element SHALL contain a unique gml:id attribute.

7.7. DualSpaceLayer

The DualSpaceLayer element is an instance of DualSpaceLayer, representing the graph structure (nodes and edges) corresponding to a primal space layer.

The DualSpaceLayer element shall contain:

- gml:id attribute
- isLogical element indicating whether the graph is logical (true) or geometric (false)
- isDirected element specifying whether the graph is directed
- nodeMember property for representing Node elements
- optionally, edgeMember property for representing Edge elements
- optionally, creationDate element
- optionally, terminationDate element

```
<element name="DualSpaceLayer" substitutionGroup="gml:AbstractFeature" type="core:
DualSpaceLayerType"/>
<complexType name="DualSpaceLayerType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element name="isLogical" type="boolean"/>
    <element minOccurs="0" name="creationDate" type="dateTime"/>
    <element minOccurs="0" name="terminationDate" type="dateTime"/>
    <element name="isDirected" type="boolean"/>
    <element maxOccurs="unbounded" minOccurs="0" name="edgeMember">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
        <sequence minOccurs="0">
         <element ref="core:Edge"/>
        </sequence>
        <attributeGroup ref="gml:AssociationAttributeGroup"/>
       </extension>
      </complexContent>
     </complexType>
    </element>
    <element maxOccurs="unbounded" name="nodeMember">
     <complexType>
      <complexContent>
       <extension base="gml:AbstractFeatureMemberType">
```

```
            <sequence minOccurs="0">
              <element ref="core:Node"/>
            </sequence>
            <attributeGroup ref="gml:AssociationAttributeGroup"/>
          </extension>
        </complexContent>
      </complexType>
    </element>
  </sequence>
 </extension>
</complexContent>
</complexType>
<complexType name="DualSpaceLayerPropertyType">
 <sequence minOccurs="0">
  <element ref="core:DualSpaceLayer"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the DualSpaceLayer element

```
<core:DualSpaceLayer gml:id="DSL_1">
   <!-- mandatory properties -->
   <core:isLogical>false</core:isLogical>
   <core:isDirected>false</core:isDirected>
   <core:nodeMember>
      <core:Node gml:id="ND_1">...</core:Node>
   </core:nodeMember>

   <!-- optional properties -->
   <core:edgeMember>
      <core:Edge gml:id="EDG_1">...</core:Edge>
   </core:edgeMember>
   <core:creationDate>2025-08-06T18:39:00Z</core:creationDate>
</core:DualSpaceLayer>
```

Listing — An example of the DualSpaceLayer element

REQUIREMENT 7: REQUIREMENT FOR DUALSPACELAYER XML ELEMENT	
IDENTIFIER	/req/core/dualspacelayer
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A DualSpaceLayer element SHALL specify whether the graph is logical (true) or geometric (false) using the isLogical element.
B	When isLogical is false, the geometries of all Node instances contained in nodeMember SHALL be spatially located inside their corresponding CellSpace geometries.
C	A DualSpaceLayer element SHALL specify whether the graph is directed using the isDirected element (true or false).
D	A DualSpaceLayer element SHALL contain at least one nodeMember property containing or referencing a Node element.

REQUIREMENT 7: REQUIREMENT FOR DUALSPACELAYER XML ELEMENT	
E	A DualSpaceLayer element SHALL contain a unique gml:id attribute.

7.8. Node

The Node element is an instance of Node, representing a state or graph vertex in the dual space.

The Node element shall contain:

- gml:id attribute
- duality property referencing the corresponding CellSpace element
- optionally, geometry element representing the position of the node
- optionally, connects property referencing connected Edge elements

```
<element name="Node" substitutionGroup="gml:AbstractFeature" type="core:NodeType"/>
<complexType name="NodeType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
    <element name="duality" type="core:CellSpacePropertyType"/>
    <element maxOccurs="unbounded" minOccurs="0" name="connects" type="core:
EdgePropertyType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="NodePropertyType">
 <sequence minOccurs="0">
  <element ref="core:Node"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the Node element

```
<core:Node gml:id="ND_1">
   <!-- mandatory property -->
   <core:duality xlink:href="#CS_1"/>

   <!-- optional properties -->
   <core:geometry>
      <gml:Point gml:id="G_ND_1">
         <gml:pos>445536.499779417 5444906.24858758 -2.02</gml:pos>
      </gml:Point>
   </core:geometry>
   <core:connects xlink:href="#EDG_1"/>
```

```
</core:Node>
```

Listing — An example of the Node element

REQUIREMENT 8: REQUIREMENT FOR NODE XML ELEMENT	
IDENTIFIER	/req/core/node
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	A Node element SHALL contain a duality property referencing (via xlink:href) or containing the corresponding CellSpace element in the primal space.
B	When present, each connects property of a Node element SHALL reference (via xlink:href) or contain a connected Edge element.
C	A Node element SHALL contain a unique gml:id attribute.

7.9. Edge

The Edge element is an instance of Edge, which represents an adjacency, connectivity, or navigability relationship between two nodes in dual space.

The Edge element shall contain:

- gml:id attribute
- weight element which can be used in graph-based applications
- connects property referencing exactly two connected Node elements
- optionally, duality property referencing the corresponding CellBoundary element
- optionally, geometry element representing the curve of the edge

```
<element name="Edge" substitutionGroup="gml:AbstractFeature" type="core:EdgeType"/>
<complexType name="EdgeType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
    <element name="weight" type="double"/>
    <element minOccurs="0" name="duality" type="core:CellBoundaryPropertyType"/>
    <element maxOccurs="2" minOccurs="2" name="connects" type="core:NodePropertyType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
```

```
<complexType name="EdgePropertyType">
  <sequence minOccurs="0">
    <element ref="core:Edge"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the Edge element

```
<core:Edge gml:id="EDG_1">
    <!-- mandatory properties -->
    <core:weight>1.0</core:weight>
    <core:connects xlink:href="#ND_1"/>
    <core:connects xlink:href="#ND_2"/>

    <!-- optional properties -->
    <core:geometry>
      <gml:LineString gml:id="G_EDG_1">
        <gml:pos>445536.499779417 5444906.24858758 -2.02</gml:pos>
        <gml:pos>445537.914644321 5444904.59001251 -2.02</gml:pos>
      </gml:LineString>
    </core:geometry>
    <core:duality xlink:href="#CB_1"/>
</core:Edge>
```

Listing — An example of the Edge element

REQUIREMENT 9: REQUIREMENT FOR EDGE XML ELEMENT	
IDENTIFIER	/req/core/edge
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	An Edge element SHALL contain exactly two connects properties referencing (via xlink:href) or containing the connected Node elements.
B	If isDirected is true in the parent DualSpaceLayer, the sequence of connects properties SHALL define the direction of the edge (from the first node to the second node).
C	An Edge element SHALL contain a weight element of type xs:double.
D	When present, the curve in geometry SHALL not self-intersect, and its start and end coordinates SHALL match the coordinates of the connected Node instances referenced by connects.
E	When present, the duality property of an Edge element SHALL reference (via xlink:href) or contain the corresponding CellBoundary element.
F	An Edge element SHALL contain a unique gml:id attribute.

7.10. InterLayerConnection

The InterLayerConnection element is an instance of InterLayerConnection, which describes a topological relationship between objects in different thematic layers.

The InterLayerConnection element shall contain:

- gml:id attribute
- connectedLayers property mapped to exactly two ThematicLayer elements
- typeOfTopoExpression element specifying the topological relationship
- optionally, connectedNodes property mapped to two Node elements
- optionally, connectedCells property mapped to two CellSpace elements
- optionally, comment element

The typeOfTopoExpression element represents the topological relationship between connected elements across different thematic layers, with values including "Contains", "Overlaps", "Equals", "Within", "Crosses", and "Other". These topological values are conceptualized as verbs in a Subject-Verb-Object (SVO) sentence structure:

- When connectedCells (or connectedNodes) are specified, the topological relationship primarily applies to those specific cell spaces (or nodes). That is, the first referenced cell or node acts as the Subject, and the second referenced cell or node acts as the Object (e.g., *Cell A of Layer 1 Contains Cell B of Layer 2*, even if Layer 1 as a whole does not fully contain Layer 2).
- In the absence of connectedCells and connectedNodes, the topological relationship applies directly between the two whole thematic layers, where the first referenced layer in connectedLayers is the Subject and the second is the Object (e.g., *Layer 1 Contains Layer 2*).

Consequently, the sequence of elements across connectedLayers, connectedCells, and connectedNodes properties shall strictly maintain this pairwise SVO order, where index 1 always represents the Subject (or the element from the Subject layer) and index 2 represents the Object (or the element from the Object layer).

```
<element name="InterLayerConnection" substitutionGroup="gml:AbstractFeature" type="core:
InterLayerConnectionType"/>
<complexType name="InterLayerConnectionType">
  <complexContent>
    <extension base="gml:AbstractFeatureType">
      <sequence>
        <element name="typeOfTopoExpression" type="core:TopoExpressionValueType"/>
        <element minOccurs="0" name="comment" type="string"/>
        <element maxOccurs="2" minOccurs="2" name="connectedLayers" type="core:
ThematicLayerPropertyType"/>
        <element maxOccurs="2" minOccurs="0" name="connectedCells" type="core:
CellSpacePropertyType"/>
```

```
      <element maxOccurs="2" minOccurs="0" name="connectedNodes" type="core:NodePropertyType"/
>
    </sequence>
   </extension>
  </complexContent>
</complexType>
<complexType name="InterLayerConnectionPropertyType">
  <sequence minOccurs="0">
    <element ref="core:InterLayerConnection"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="TopoExpressionValueType">
  <union memberTypes="core:TopoExpressionValueEnumerationType core:
TopoExpressionValueOtherType"/>
</simpleType>
<simpleType name="TopoExpressionValueEnumerationType">
  <restriction base="string">
    <enumeration value="Contains"/>
    <enumeration value="Overlaps"/>
    <enumeration value="Equals"/>
    <enumeration value="Within"/>
    <enumeration value="Crosses"/>
    <enumeration value="Other"/>
  </restriction>
</simpleType>
<simpleType name="TopoExpressionValueOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the InterLayerConnection element

```
<core:InterLayerConnection gml:id="ILC_1">
    <!-- mandatory properties -->
    <core:typeOfTopoExpression>Contains</core:typeOfTopoExpression>
    <core:connectedLayers xlink:href="#TL_Pedestrian"/>
    <core:connectedLayers xlink:href="#TL_Wheelchair"/>

    <!-- optional properties -->
    <core:comment>Pedestrian layer contains wheelchair layer subdivision</core:comment>
    <core:connectedCells xlink:href="#CS_Ped_1"/>
    <core:connectedCells xlink:href="#CS_Wheel_1"/>
</core:InterLayerConnection>
```

Listing — An example of the InterLayerConnection element

REQUIREMENT 10: REQUIREMENT FOR INTERLAYERCONNECTION XML ELEMENT	
IDENTIFIER	/req/core/interlayerconnection
INCLUDED IN	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
A	An InterLayerConnection element SHALL contain exactly two connectedLayers properties referencing (via xlink:href) or containing two distinct ThematicLayer elements.

REQUIREMENT 10: REQUIREMENT FOR INTERLAYERCONNECTION XML ELEMENT	
B	An InterLayerConnection element SHALL specify the topological relationship using the typeOfTopoExpression element, which applies primarily to the referenced connectedCells (or connectedNodes), or to the connectedLayers when no cells or nodes are referenced.
C	The typeOfTopoExpression value SHALL be one of the following: "Contains", "Overlaps", "Equals", "Within", "Crosses", "Other", or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: Touches").
D	When present, connectedCells SHALL contain exactly two properties referencing (via xlink:href) or containing CellSpace elements belonging to the primal space layers of the first and second connected thematic layers, respectively.
E	When present, connectedNodes SHALL contain exactly two properties referencing (via xlink:href) or containing Node elements belonging to the dual space layers of the first and second connected thematic layers, respectively.
F	The sequence of elements in the connectedLayers, connectedCells, and connectedNodes properties SHALL be ordered such that the first element always represents the subject of the topological relationship, and the second element represents the object, ensuring they pairwise correspond at the same indices.
G	An InterLayerConnection element SHALL contain a unique gml:id attribute.

7.11. ExternalReference

The ExternalReference element is used for referencing an object in an external dataset from an IndoorGML feature.

```
<element name="ExternalReference" substitutionGroup="gml:AbstractObject" type="core:
ExternalReferenceType"/>
<complexType name="ExternalReferenceType">
 <sequence>
  <element name="externalObject" type="core:ExternalObjectReferencePropertyType"/>
  <element name="informationSystem" type="gmd:URL_PropertyType"/>
 </sequence>
</complexType>
<element name="ExternalObjectReference" substitutionGroup="gml:AbstractObject" type="core:
ExternalObjectReferenceType"/>
<complexType name="ExternalObjectReferenceType">
 <sequence>
  <element name="name" type="string"/>
  <element name="uri" type="anyURI"/>
 </sequence>
</complexType>
```

Listing — XML schema for the ExternalReference element

```
<core:externalReference>
  <core:ExternalReference>
```

```
      <core:externalObject>
        <core:ExternalObjectReference>
          <core:name>IFC_Space_101</core:name>
          <core:uri>urn:uuid:2b88523c-a9e3-4c92-9442-5f6966f3630f</core:uri>
        </core:ExternalObjectReference>
      </core:externalObject>
      <core:informationSystem>
        <gmd:URL>https://bim.example.org/project/buildingA.ifc</gmd:URL>
      </core:informationSystem>
    </core:ExternalReference>
  </core:externalReference>
```

Listing — An example of the ExternalReference element

8

INDOORXML NAVIGATION

8 INDOORXML NAVIGATION

REQUIREMENTS CLASS 2	
IDENTIFIER	http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
TARGET TYPE	Implementation Specification
CONFORMANCE CLASS	Conformance class A.2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi
PREREQUISITES	http://www.opengis.net/spec/indoorgml/2.0/req Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
NORMATIVE STATEMENTS	Requirement 11: /req/navi/global Requirement 12: /req/navi/navigablespace Requirement 13: /req/navi/generalspace Requirement 14: /req/navi/transferspace Requirement 15: /req/navi/navigableboundary Requirement 16: /req/navi/nonnavigablespace Requirement 17: /req/navi/objectspace Requirement 18: /req/navi/nonnavigableboundary Requirement 19: /req/navi/route

8.1. General requirements for encoding standards

The IndoorGML Navigation extension module provides semantic information and specialized feature types to support indoor navigation applications. It extends the Core module by classifying indoor spaces and boundaries into navigable and non-navigable categories, and by providing representations for designated navigation routes.

An instance XML document representing an IndoorGML 2.0 Navigation module dataset SHALL be well-formed XML and conform to the XML schemas defined in this standard (IndoorGML_navi.xsd, IndoorGML_core.xsd) and OGC GML 3.2.1.

REQUIREMENT 11: REQUIREMENT FOR NAVIGATION MODULE XML SCHEMA VALIDATION	
IDENTIFIER	/req/navi/global
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi

REQUIREMENT 11: REQUIREMENT FOR NAVIGATION MODULE XML SCHEMA VALIDATION	
A	An instance XML document representing an IndoorGML 2.0 Navigation module dataset SHALL be well-formed XML and validate against the schemas defined in this standard (including IndoorGML_navi.xsd, IndoorGML_core.xsd, and OGC GML 3.2.1).

8.2. NavigableSpace

The NavigableSpace element is an instance of NavigableSpace, representing a space where an agent or user can move freely (e.g. corridor, door, lobby, hallway, room).

The NavigableSpace element shall contain:

- all properties inherited from the CellSpace element
- locomotionType element indicating the allowed locomotion mode in indoor space

```
<element name="NavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NavigableSpaceType"/>
<complexType name="NavigableSpaceType">
 <complexContent>
  <extension base="core:CellSpaceType">
   <sequence>
    <element name="locomotionType" type="navi:LocomotionAccessType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="NavigableSpacePropertyType">
 <sequence minOccurs="0">
  <element ref="navi:NavigableSpace"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="LocomotionAccessType">
 <union memberTypes="navi:LocomotionAccessEnumerationType navi:LocomotionAccessOtherType"/>
</simpleType>
<simpleType name="LocomotionAccessEnumerationType">
 <restriction base="string">
  <enumeration value="Walking"/>
  <enumeration value="Flying"/>
  <enumeration value="Rolling"/>
  <enumeration value="Unspecified"/>
 </restriction>
</simpleType>
<simpleType name="LocomotionAccessOtherType">
 <restriction base="string">
  <pattern value="other: \w{2,}"/>
 </restriction>
```

```
</simpleType>
```

Listing — XML schema for the NavigableSpace element

```
<navi:NavigableSpace gml:id="NS_1">
   <!-- inherited mandatory property from CellSpace -->
   <core:PoI>false</core:PoI>

   <!-- mandatory property -->
   <navi:locomotionType>Walking</navi:locomotionType>
</navi:NavigableSpace>
```

Listing — An example of the NavigableSpace element

REQUIREMENT 12: REQUIREMENT FOR NAVIGABLESPACE XML ELEMENT	
IDENTIFIER	/req/navi/navigablespace
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A NavigableSpace element SHALL conform to all schema and semantic requirements of Cell Space.
B	A NavigableSpace element SHALL specify the allowed locomotion mode using the locomotionType element.
C	The locomotionType value SHALL be one of the following: "Walking", "Flying", "Rolling", "Unspecified", or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: Drone" or "other: Segway").
D	A NavigableSpace element SHALL contain a unique gml:id attribute.

8.3. GeneralSpace

The GeneralSpace element is an instance of GeneralSpace, representing a navigable space that agents can occupy for extended periods and which can serve as starting or destination points in navigation.

The GeneralSpace element shall contain:

- all properties inherited from the NavigableSpace element
- function element specifying details about the functional purpose of the space

```
<element name="GeneralSpace" substitutionGroup="navi:NavigableSpace" type="navi:
GeneralSpaceType"/>
<complexType name="GeneralSpaceType">
  <complexContent>
```

```
    <extension base="navi:NavigableSpaceType">
     <sequence>
      <element name="function" type="navi:GeneralSpaceFunctionType"/>
     </sequence>
    </extension>
   </complexContent>
</complexType>
<complexType name="GeneralSpacePropertyType">
  <sequence minOccurs="0">
    <element ref="navi:GeneralSpace"/>
  </sequence>
  <attributeGroup ref="gml:AssociationAttributeGroup"/>
  <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="GeneralSpaceFunctionType">
  <union memberTypes="navi:GeneralSpaceFunctionEnumerationType navi:
GeneralSpaceFunctionOtherType"/>
</simpleType>
<simpleType name="GeneralSpaceFunctionEnumerationType">
  <restriction base="string">
    <enumeration value="Administration"/>
    <enumeration value="Business, trade"/>
    <enumeration value="Education, training"/>
    <enumeration value="Recreation"/>
    <enumeration value="Laboratory"/>
    <enumeration value="Storage"/>
    <enumeration value="Security"/>
  </restriction>
</simpleType>
<simpleType name="GeneralSpaceFunctionOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the GeneralSpace element

```
<navi:GeneralSpace gml:id="GS_1">
    <!-- inherited mandatory properties -->
    <core:PoI>false</core:PoI>
    <navi:locomotionType>Walking</navi:locomotionType>

    <!-- mandatory property -->
    <navi:function>Storage</navi:function>
</navi:GeneralSpace>
```

Listing — An example of the GeneralSpace element

REQUIREMENT 13: REQUIREMENT FOR GENERALSPACE XML ELEMENT	
IDENTIFIER	/req/navi/generalspace
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A GeneralSpace element SHALL conform to all schema and semantic requirements of NavigableSpace.

REQUIREMENT 13: REQUIREMENT FOR GENERALSPACE XML ELEMENT	
B	A GeneralSpace element SHALL specify the functional purpose of the space using the function element.
C	The function value SHALL be one of the following: "Administration", "Business, trade", "Education, training", "Recreation", "Laboratory", "Storage", "Security", or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: Cafeteria").
D	A GeneralSpace element SHALL contain a unique gml:id attribute.

8.4. TransferSpace

The TransferSpace element is an instance of TransferSpace, representing a navigable space that provides passage between general spaces (e.g. doors, windows, gates).

The TransferSpace element shall contain:

- all properties inherited from the NavigableSpace element
- function element describing whether the space is an “AnchorSpace” or a “ConnectionSpace”
- type element specifying opening categories (e.g., Door or Window)

Note that an AnchorSpace is a space that allows for the connection between the indoor and the outdoor. A ConnectionSpace is a space connecting two indoor or two outdoor spaces.

```
<element name="TransferSpace" substitutionGroup="navi:NavigableSpace" type="navi:
TransferSpaceType"/>
<complexType name="TransferSpaceType">
 <complexContent>
  <extension base="navi:NavigableSpaceType">
   <sequence>
    <element name="function" type="navi:TransferSpaceFunctionType"/>
    <element name="type" type="navi:TransferSpaceTypeType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="TransferSpacePropertyType">
 <sequence minOccurs="0">
  <element ref="navi:TransferSpace"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="TransferSpaceFunctionType">
 <union memberTypes="navi:TransferSpaceFunctionEnumerationType navi:TransferSpaceFunctionOthe
rType"/>
</simpleType>
```

```
<simpleType name="TransferSpaceFunctionEnumerationType">
 <restriction base="string">
  <enumeration value="ConnectionSpace"/>
  <enumeration value="AnchorSpace"/>
 </restriction>
</simpleType>
<simpleType name="TransferSpaceFunctionOtherType">
 <restriction base="string">
  <pattern value="other: \w{2,}"/>
 </restriction>
</simpleType>
<simpleType name="TransferSpaceTypeType">
 <union memberTypes="navi:TransferSpaceTypeEnumerationType navi:TransferSpaceTypeOtherType"/
>
</simpleType>
<simpleType name="TransferSpaceTypeEnumerationType">
 <restriction base="string">
  <enumeration value="Door"/>
  <enumeration value="Window"/>
 </restriction>
</simpleType>
<simpleType name="TransferSpaceTypeOtherType">
 <restriction base="string">
  <pattern value="other: \w{2,}"/>
 </restriction>
</simpleType>
```

Listing — XML schema for the TransferSpace element

```
<navi:TransferSpace gml:id="TS_1">
   <!-- inherited mandatory properties -->
   <core:PoI>false</core:PoI>
   <navi:locomotionType>Walking</navi:locomotionType>

   <!-- mandatory properties -->
   <navi:function>ConnectionSpace</navi:function>
   <navi:type>Door</navi:type>
</navi:TransferSpace>
```

Listing — An example of the TransferSpace element

REQUIREMENT 14: REQUIREMENT FOR TRANSFERSPACE XML ELEMENT	
IDENTIFIER	/req/navi/transferspace
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A TransferSpace element SHALL conform to all schema and semantic requirements of NavigableSpace.
B	A TransferSpace element SHALL specify its passage function using the function element and its opening category using the type element.
C	The function value SHALL be one of the following: "ConnectionSpace" (connecting two indoor or two outdoor spaces), "AnchorSpace" (enabling connection between indoor and outdoor spaces), or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: InterbuildingSpace").

REQUIREMENT 14: REQUIREMENT FOR TRANSFERSPACE XML ELEMENT	
D	The type value SHALL be one of the following: "Door", "Window", or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: Turnstile" or "other: Elevator").
E	A TransferSpace element SHALL contain a unique gml:id attribute.

8.5. NavigableBoundary

The NavigableBoundary element is an instance of NavigableBoundary, representing a boundary of a navigable space that provides transition or connection information.

The NavigableBoundary element shall contain:

- all properties inherited from the CellBoundary element
- NavigableBoundaryFunction element describing whether the boundary is an “AnchorBoundary” or a “ConnectionBoundary”
- optionally, boundaryOrientation element indicating whether to use the geometry represented by cellBoundaryGeom as-is (true) or in reverse (false)

Note that an AnchorBoundary is the boundary of an AnchorSpace that enables the connection between indoor and outdoor spaces. A ConnectionBoundary is the boundary of a ConnectionSpace that connects two indoor spaces or two outdoor spaces.

```
<element name="NavigableBoundary" substitutionGroup="core:CellBoundary" type="navi:
NavigableBoundaryType"/>
<complexType name="NavigableBoundaryType">
 <complexContent>
  <extension base="core:CellBoundaryType">
   <sequence>
    <element name="NavigableBoundaryFunction" type="navi:NavigableBoundaryFunctionType"/>
    <element minOccurs="0" name="boundaryOrientation" type="boolean"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="NavigableBoundaryPropertyType">
 <sequence minOccurs="0">
  <element ref="navi:NavigableBoundary"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="NavigableBoundaryFunctionType">
 <union memberTypes="navi:NavigableBoundaryFunctionEnumerationType navi:NavigableBoundaryFu
nctionOtherType"/>
</simpleType>
<simpleType name="NavigableBoundaryFunctionEnumerationType">
```

```
  <restriction base="string">
    <enumeration value="AncorBoundary"/>
    <enumeration value="ConnectionBoundary"/>
  </restriction>
</simpleType>
<simpleType name="NavigableBoundaryFunctionOtherType">
  <restriction base="string">
    <pattern value="other: \w{2,}"/>
  </restriction>
</simpleType>
```

Listing — XML schema for the NavigableBoundary element

```
<navi:NavigableBoundary gml:id="NB_1">
    <!-- inherited mandatory property from CellBoundary -->
    <core:isVirtual>false</core:isVirtual>

    <!-- mandatory property -->
    <navi:NavigableBoundaryFunction>ConnectionBoundary</navi:NavigableBoundaryFunction>

    <!-- optional property -->
    <navi:boundaryOrientation>true</navi:boundaryOrientation>
</navi:NavigableBoundary>
```

Listing — An example of the NavigableBoundary element

REQUIREMENT 15: REQUIREMENT FOR NAVIGABLEBOUNDARY XML ELEMENT	
IDENTIFIER	/req/navi/navigableboundary
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A NavigableBoundary element SHALL conform to all schema and semantic requirements of Cell Boundary.
B	A NavigableBoundary element SHALL specify its boundary function using the NavigableBoundaryFunction element.
C	The NavigableBoundaryFunction value SHALL be one of the following: "ConnectionBoundary" (boundary connecting two indoor or two outdoor spaces), "AncorBoundary" (boundary enabling transition between indoor and outdoor spaces), or a user-defined value matching the pattern "other: \w{2,}" (e.g., "other: VirtualBoundary").
D	When present, boundaryOrientation SHALL specify whether the boundary geometry is used with standard orientation (true) or reversed orientation (false).
E	A NavigableBoundary element SHALL contain a unique gml:id attribute.

8.6. NonNavigableSpace

The NonNavigableSpace element is an instance of NonNavigableSpace, representing an indoor space subdivision occupied by obstacles or restricted from agent traversal.

The NonNavigableSpace element shall contain:

- all properties inherited from the CellSpace element

```
<element name="NonNavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NonNavigableSpaceType"/>
<complexType name="NonNavigableSpaceType">
 <complexContent>
  <extension base="core:CellSpaceType">
   <sequence/>
  </extension>
 </complexContent>
</complexType>
<complexType name="NonNavigableSpacePropertyType">
 <sequence minOccurs="0">
  <element ref="navi:NonNavigableSpace"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the NonNavigableSpace element

```
<navi:NonNavigableSpace gml:id="NNS_1">
  <!-- inherited mandatory property from CellSpace -->
  <core:PoI>false</core:PoI>
</navi:NonNavigableSpace>
```

Listing — An example of the NonNavigableSpace element

REQUIREMENT 16: REQUIREMENT FOR NONNAVIGABLESPACE XML ELEMENT	
IDENTIFIER	/req/navi/nonnavigablespace
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A NonNavigableSpace element SHALL conform to all schema and semantic requirements of CellSpace.
B	A NonNavigableSpace element SHALL contain a unique gml:id attribute.

8.7. ObjectSpace

The ObjectSpace element is an instance of ObjectSpace, representing a specialized non-navigable space that encapsulates physical objects or furniture within an indoor environment.

The ObjectSpace element shall contain:

- all properties inherited from the NonNavigableSpace element
- optionally, description element providing descriptive text regarding the contained objects
- optionally, containedFeatures element representing the number of objects encapsulated within the ObjectSpace

```
<element name="ObjectSpace" substitutionGroup="navi:NonNavigableSpace" type="navi:
ObjectSpaceType"/>
<complexType name="ObjectSpaceType">
 <complexContent>
  <extension base="navi:NonNavigableSpaceType">
   <sequence>
    <element minOccurs="0" name="description" type="string"/>
    <element minOccurs="0" name="containedFeatures" type="integer"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="ObjectSpacePropertyType">
 <sequence minOccurs="0">
  <element ref="navi:ObjectSpace"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the ObjectSpace element

```
<navi:ObjectSpace gml:id="OS_1">
  <!-- inherited mandatory property from CellSpace -->
  <core:PoI>false</core:PoI>

  <!-- optional properties -->
  <navi:containedFeatures>3</navi:containedFeatures>
  <navi:description>Living room furniture and columns</navi:description>
</navi:ObjectSpace>
```

Listing — An example of the ObjectSpace element

REQUIREMENT 17: REQUIREMENT FOR OBJECTSPACE XML ELEMENT	
IDENTIFIER	/req/navi/objectspace
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi

REQUIREMENT 17: REQUIREMENT FOR OBJECTSPACE XML ELEMENT	
A	An ObjectSpace element SHALL conform to all schema and semantic requirements of NonNavigableSpace.
B	An ObjectSpace element SHALL NOT overlap with any other CellSpace or its specialized classes within the same PrimalSpaceLayer.
C	When present, containedFeatures SHALL be a non-negative integer representing the count of contained objects.
D	When present, description SHALL provide descriptive text regarding the contained objects.
E	An ObjectSpace element SHALL contain a unique gml:id attribute.

8.8. NonNavigableBoundary

The NonNavigableBoundary element is an instance of NonNavigableBoundary, representing the boundary between two non-navigable spaces or between a navigable space and a non-navigable space.

The NonNavigableBoundary element shall contain:

- all properties inherited from the CellBoundary element

```
<element name="NonNavigableBoundary" substitutionGroup="core:CellBoundary" type="navi:
NonNavigableBoundaryType"/>
<complexType name="NonNavigableBoundaryType">
 <complexContent>
  <extension base="core:CellBoundaryType">
   <sequence/>
  </extension>
 </complexContent>
</complexType>
<complexType name="NonNavigableBoundaryPropertyType">
 <sequence minOccurs="0">
  <element ref="navi:NonNavigableBoundary"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
```

Listing — XML schema for the NonNavigableBoundary element

```
<navi:NonNavigableBoundary gml:id="NNB_1">
  <!-- inherited mandatory property from CellBoundary -->
  <core:isVirtual>false</core:isVirtual>
</navi:NonNavigableBoundary>
```

Listing — An example of the NonNavigableBoundary element

REQUIREMENT 18: REQUIREMENT FOR NONNAVIGABLEBOUNDARY XML ELEMENT	
IDENTIFIER	/req/navi/nonnavigableboundary
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A NonNavigableBoundary element SHALL conform to all schema and semantic requirements of CellBoundary.
B	A NonNavigableBoundary element SHALL contain a unique gml:id attribute.

8.9. Route

The Route element is an instance of Route, representing a designated navigation path through an indoor space composed of an ordered sequence of nodes and connecting edges in dual space.

The Route element shall contain:

- gml:id attribute
- routeNode properties containing ordered sequences of Node references to describe the different parts of the route path
- routeEdge properties containing ordered sequences of Edge references to describe the different parts of the route path
- optionally, creationDate element indicating when a specific route was created

```
<element name="Route" substitutionGroup="gml:AbstractFeature" type="navi:RouteType"/>
<complexType name="RouteType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element minOccurs="0" name="creationDate" type="dateTime"/>
    <element maxOccurs="unbounded" minOccurs="2" name="routeNode" type="core:
NodePropertyType"/>
     <element maxOccurs="unbounded" name="routeEdge" type="core:EdgePropertyType"/>
   </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="RoutePropertyType">
 <sequence minOccurs="0">
  <element ref="navi:Route"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
```

```
</complexType>
```

Listing — XML schema for the Route element

```
<navi:Route gml:id="RT_1">
   <!-- mandatory properties (at least 2 nodes and 1 edge) -->
   <navi:routeNode xlink:href="#ND_1"/>
   <navi:routeNode xlink:href="#ND_2"/>
   <navi:routeEdge xlink:href="#EDG_1"/>

   <!-- optional property -->
   <navi:creationDate>2025-10-20T16:44:00Z</navi:creationDate>
</navi:Route>
```

Listing — An example of the Route element

REQUIREMENT 19: REQUIREMENT FOR ROUTE XML ELEMENT	
IDENTIFIER	/req/navi/route
INCLUDED IN	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
A	A Route element SHALL contain at least two routeNode properties and at least one routeEdge property.
B	Each routeNode property SHALL reference (via xlink:href) or contain a Node element.
C	Each routeEdge property SHALL reference (via xlink:href) or contain an Edge element.
D	The sequences of routeNode and routeEdge properties SHALL represent an ordered contiguous path in the dual space graph.
E	Each consecutive pair of routeNode elements in the sequence SHALL be connected by the corresponding routeEdge element in routeEdge.
F	Consecutive routeEdge elements in the sequence SHALL share a common routeNode element.
G	When present, creationDate SHALL be formatted as a valid xs:dateTime string.
H	A Route element SHALL contain a unique gml:id attribute.

A

ANNEX A (NORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE

ANNEX A (NORMATIVE) CONFORMANCE CLASS ABSTRACT TEST SUITE

A.1. Conformance Class Core

CONFORMANCE CLASS A.1	
IDENTIFIER	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/core
REQUIREMENTS CLASS	Requirements class 1: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/core
TARGET TYPE	Implementation Specification
CONFORMANCE TESTS	Abstract test A.1: /conf/core/global Abstract test A.2: /conf/core/indoorfeatures Abstract test A.3: /conf/core/thematiclayer Abstract test A.4: /conf/core/primalspacelayer Abstract test A.5: /conf/core/cellspace Abstract test A.6: /conf/core/cellboundary Abstract test A.7: /conf/core/dualspacelayer Abstract test A.8: /conf/core/node Abstract test A.9: /conf/core/edge Abstract test A.10: /conf/core/interlayerconnection

ABSTRACT TEST A.1	
IDENTIFIER	/conf/core/global
REQUIREMENT	Requirement 1: /req/core/global
TEST PURPOSE	Verify that instance XML documents validate against the IndoorGML 2.0 Core XML schema (IndoorGML_core.xsd) and OGC GML 3.2.1 schemas.

ABSTRACT TEST A.1

TEST METHOD	Validate the XML instance document against IndoorGML_core.xsd and supporting GML 3.2.1 schemas using an XML Schema validator or appropriate validation software.

ABSTRACT TEST A.2

IDENTIFIER	/conf/core/indoorfeatures
REQUIREMENT	Requirement 2: /req/core/indoorfeatures
TEST PURPOSE	Verify that an IndoorFeatures element contains at least one layers property referencing or containing a ThematicLayer element, that optional layerConnections properties contain or reference InterLayerConnection elements, and that a unique gml:id is present.
TEST METHOD	Inspect the instance document to verify that the layers property has at least one valid Thematic Layer instance, that layerConnections contain valid InterLayerConnection instances, and that gml: id is unique.

ABSTRACT TEST A.3

IDENTIFIER	/conf/core/thematiclayer
REQUIREMENT	Requirement 3: /req/core/thematiclayer
TEST PURPOSE	Verify that for each ThematicLayer element: • semanticExtension is a valid boolean value (true or false). • theme contains a valid value ("Tags", "Virtual", "Unknown", "Physical", or "other: \w{2,}" such as "other: WiFi"). • If present, primalSpace references or contains a valid PrimalSpaceLayer element. • If present, dualSpace references or contains a valid DualSpaceLayer element. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify the above properties.

ABSTRACT TEST A.4

IDENTIFIER	/conf/core/primalspacelayer
REQUIREMENT	Requirement 4: /req/core/primalspacelayer

ABSTRACT TEST A.4

TEST PURPOSE	Verify that for each PrimalSpaceLayer element: • At least one cellSpaceMember property containing or referencing a CellSpace element is present. • The geometries of all CellSpace members belonging to the same PrimalSpaceLayer do not intersect each other. • The geometries of all CellBoundary members belonging to the same PrimalSpaceLayer do not intersect each other. • If present, creationDate and terminationDate are valid xs:dateTime strings. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document and perform spatial intersection analysis on member geometries to verify non-overlapping constraints.

ABSTRACT TEST A.5

IDENTIFIER	/conf/core/cellspace
REQUIREMENT	Requirement 5: /req/core/cellspace
TEST PURPOSE	Verify that for each CellSpace element: • PoI is a valid boolean value (true or false). • If present, duality references or contains a corresponding Node element in dual space. • If present, each boundedBy references or contains a corresponding CellBoundary element. • The geometry of each bounding CellBoundary does not exceed the geometric extent of the CellSpace. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document and verify geometric boundary containment against the cell space geometry.

ABSTRACT TEST A.6

IDENTIFIER	/conf/core/cellboundary
REQUIREMENT	Requirement 6: /req/core/cellboundary
TEST PURPOSE	Verify that for each CellBoundary element: • isVirtual is a valid boolean value (true or false). • If present, duality references or contains a corresponding Edge element in dual space. • A unique gml:id attribute is present.

ABSTRACT TEST A.6

TEST METHOD	Inspect the instance document to verify the above properties.

ABSTRACT TEST A.7

IDENTIFIER	/conf/core/dualspacelayer
REQUIREMENT	Requirement 7: /req/core/dualspacelayer
TEST PURPOSE	Verify that for each DualSpaceLayer element: • isLogical and isDirected are valid boolean values (true or false). • When isLogical is false, the geometry of each Node in nodeMember is spatially located inside the geometry of its corresponding CellSpace. • At least one nodeMember property containing or referencing a Node element is present. • If present, edgeMember properties contain or reference Edge elements. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document and perform point-in-polygon / point-in-solid geometric tests when isLogical is false.

ABSTRACT TEST A.8

IDENTIFIER	/conf/core/node
REQUIREMENT	Requirement 8: /req/core/node
TEST PURPOSE	Verify that for each Node element: • duality references or contains the corresponding CellSpace element in the primal space. • If present, each connects property references or contains a connected Edge element. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify cross-references between nodes, edges, and cell spaces.

ABSTRACT TEST A.9

IDENTIFIER	/conf/core/edge
REQUIREMENT	Requirement 9: /req/core/edge

ABSTRACT TEST A.9

TEST PURPOSE	Verify that for each Edge element: • Exactly two connects properties referencing or containing connected Node elements are present. • If the parent DualSpaceLayer has isDirected = true, the order of connects is interpreted as directional (start to end node). • A valid weight value of type xs:double is present. • If geometry is present, the curve does not self-intersect, and its start and end coordinates match the positions of the connected Node instances. • If present, duality references or contains the corresponding CellBoundary element. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document and verify geometric coordinate alignment with connected node positions.

ABSTRACT TEST A.10

IDENTIFIER	/conf/core/interlayerconnection
REQUIREMENT	Requirement 10: /req/core/interlayerconnection
TEST PURPOSE	Verify that for each InterLayerConnection element: • Exactly two connectedLayers properties referencing distinct ThematicLayer elements are present. • typeOfTopoExpression is one of the valid enumeration or pattern values ("Contains", "Overlaps", "Equals", "Within", "Crosses", "Other", or "other: \w{2,}" such as "other: Touches"). • If present, connectedCells contains exactly two references to CellSpace elements belonging respectively to the first and second connected thematic layers. • If present, connectedNodes contains exactly two references to Node elements belonging respectively to the first and second connected thematic layers. • The sequence of connectedLayers, connectedCells, and connectedNodes properties is ordered such that the first element represents the subject of the topological relationship, and the second element represents the object, corresponding pairwise to the same indices. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify layer and object cross-references and topological expression values.

A.2. Conformance Class Navigation Extension

CONFORMANCE CLASS A.2	
IDENTIFIER	http://www.opengis.net/spec/indoorgml-2/2.0/xml/conf/navi
REQUIREMENTS CLASS	Requirements class 2: http://www.opengis.net/spec/indoorgml-2/2.0/xml/req/navi
TARGET TYPE	Implementation Specification
CONFORMANCE TESTS	Abstract test A.11: /conf/navi/global Abstract test A.12: /conf/navi/navigablespace Abstract test A.13: /conf/navi/generalspace Abstract test A.14: /conf/navi/transferspace Abstract test A.15: /conf/navi/navigableboundary Abstract test A.16: /conf/navi/nonnavigablespace Abstract test A.17: /conf/navi/objectspace Abstract test A.18: /conf/navi/nonnavigableboundary Abstract test A.19: /conf/navi/route

ABSTRACT TEST A.11	
IDENTIFIER	/conf/navi/global
REQUIREMENT	Requirement 11: /req/navi/global
TEST PURPOSE	Verify that instance XML documents using the Navigation module validate against IndoorGML_navi.xsd, IndoorGML_core.xsd, and OGC GML 3.2.1 schemas.
TEST METHOD	Validate the XML instance document against the schemas using an XML Schema validator or appropriate validation software.

ABSTRACT TEST A.12	
IDENTIFIER	/conf/navi/navigablespace
REQUIREMENT	Requirement 12: /req/navi/navigablespace
TEST PURPOSE	Verify that for each NavigableSpace element: • All validation criteria for /conf/core/cellspace are satisfied.

ABSTRACT TEST A.12

	• locomotionType contains a valid value ("Walking", "Flying", "Rolling", "Unspecified", or "other: \w{2,}" such as "other: Drone"). • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify locomotionType and inherited CellSpace properties.

ABSTRACT TEST A.13

IDENTIFIER	/conf/navi/generalspace
REQUIREMENT	Requirement 13: /req/navi/generalspace
TEST PURPOSE	Verify that for each GeneralSpace element: • All validation criteria for /conf/navi/navigablespace are satisfied. • function contains a valid value ("Administration", "Business, trade", "Education, training", "Recreation", "Laboratory", "Storage", "Security", or "other: \w{2,}" such as "other: Cafeteria"). • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify function type and inherited properties.

ABSTRACT TEST A.14

IDENTIFIER	/conf/navi/transferspace
REQUIREMENT	Requirement 14: /req/navi/transferspace
TEST PURPOSE	Verify that for each TransferSpace element: • All validation criteria for /conf/navi/navigablespace are satisfied. • function contains a valid value ("ConnectionSpace", "AnchorSpace", or "other: \w{2,}" such as "other: InterbuildingSpace"). • type contains a valid value ("Door", "Window", or "other: \w{2,}" such as "other: Turnstile"). • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify function, opening type, and inherited properties.

ABSTRACT TEST A.15

IDENTIFIER	/conf/navi/navigableboundary

ABSTRACT TEST A.15

REQUIREMENT	Requirement 15: /req/navi/navigableboundary
TEST PURPOSE	Verify that for each NavigableBoundary element: • All validation criteria for /conf/core/cellboundary are satisfied. • NavigableBoundaryFunction contains a valid value ("ConnectionBoundary", "AncorBoundary" / "AnchorBoundary", or "other: \w{2,}" such as "other: VirtualBoundary"). • If present, boundaryOrientation is a valid boolean value (true or false). • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify boundary function and orientation.

ABSTRACT TEST A.16

IDENTIFIER	/conf/navi/nonnavigablespace
REQUIREMENT	Requirement 16: /req/navi/nonnavigablespace
TEST PURPOSE	Verify that for each NonNavigableSpace element: • All validation criteria for /conf/core/cellspace are satisfied. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify conformity with CellSpace requirements.

ABSTRACT TEST A.17

IDENTIFIER	/conf/navi/objectspace
REQUIREMENT	Requirement 17: /req/navi/objectspace
TEST PURPOSE	Verify that for each ObjectSpace element: • All validation criteria for /conf/navi/nonnavigablespace are satisfied. • The geometry does not overlap with any other CellSpace or its specialized classes in the same PrimalSpaceLayer. • If present, containedFeatures is a valid non-negative integer. • If present, description is a valid string. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document and verify non-overlapping geometric relationships in the primal space.

ABSTRACT TEST A.18

IDENTIFIER	/conf/navi/nonnavigableboundary
REQUIREMENT	Requirement 18: /req/navi/nonnavigableboundary
TEST PURPOSE	Verify that for each NonNavigableBoundary element: • All validation criteria for /conf/core/cellboundary are satisfied. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document to verify conformity with CellBoundary requirements.

ABSTRACT TEST A.19

IDENTIFIER	/conf/navi/route
REQUIREMENT	Requirement 19: /req/navi/route
TEST PURPOSE	Verify that for each Route element: • At least two routeNode properties and at least one routeEdge property are present. • Each routeNode references or contains a Node element within the same dual space. • Each routeEdge references or contains an Edge element within the same dual space. • routeNode and routeEdge sequences represent an ordered contiguous path where each consecutive pair of nodes is connected by the corresponding edge, and consecutive edges share a common node. • If present, creationDate is a valid xs:dateTime string. • A unique gml:id attribute is present.
TEST METHOD	Inspect the instance document and perform graph continuity validation along the route path.

B

ANNEX B (NORMATIVE) XML SCHEMA FOR INDOORGML 2

ANNEX B
(NORMATIVE)
XML SCHEMA FOR INDOORGML 2

In addition to this document, this standard includes some normative XML Schema Documents. These XML Schema Documents are included in this document. These XML Schema Documents are also posted online at the URL http://schemas.opengis.net/indoorgml/2.0. In the event of a discrepancy between this document and online versions of the XML Schema Document files (posted at schemas.opengis.net), the online schema files SHALL be considered authoritative.

B.1. IndoorGML Core Module

```
<?xml version="1.0" encoding="UTF-8"?>
<schema
   xmlns="http://www.w3.org/2001/XMLSchema"
 xmlns:core="http://www.opengis.net/indoorgml/2.0/core"
 xmlns:gmd="http://www.isotc211.org/2005/gmd"
 xmlns:gml="http://www.opengis.net/gml/3.2"
 elementFormDefault="qualified"
 targetNamespace="http://www.opengis.net/indoorgml/2.0/core"
 version="2.0">
  <annotation>
   <appinfo>
    <gml:gmlProfileSchema>IndoorGML</gml:gmlProfileSchema>
   </appinfo>
  </annotation>
  <import namespace="http://www.isotc211.org/2005/gmd" schemaLocation="http://schemas.opengis.
net/iso/19139/20070417/gmd/gmd.xsd"/>
  <import namespace="http://www.opengis.net/gml/3.2" schemaLocation="http://schemas.opengis.net/
gml/3.2.1/gml.xsd"/>
  <!--XML Schema document created by ShapeChange - http://shapechange.net/-->
  <element name="CellBoundary" substitutionGroup="gml:AbstractFeature" type="core:
CellBoundaryType"/>
  <complexType name="CellBoundaryType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element minOccurs="0" name="cellBoundaryGeom" type="core:CellBoundaryGeometryPropertyT
ype"/>
      <element minOccurs="0" name="externalReference" type="core:ExternalReferencePropertyType"/>
      <element name="isVirtual" type="boolean"/>
      <element minOccurs="0" name="duality" type="core:EdgePropertyType">
       <annotation>
        <documentation>Corresponding cell boundary in the primal space</documentation>
        <appinfo>
         <gml:reversePropertyName>core:duality</gml:reversePropertyName>
        </appinfo>
       </annotation>
      </element>
```

```
      </sequence>
     </extension>
    </complexContent>
   </complexType>
   <complexType name="CellBoundaryPropertyType">
    <sequence minOccurs="0">
     <element ref="core:CellBoundary"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
   </complexType>
   <element name="CellBoundaryGeometry" substitutionGroup="gml:AbstractObject" type="core:
CellBoundaryGeometryType"/>
   <complexType name="CellBoundaryGeometryType">
    <sequence>
     <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
     <element minOccurs="0" name="Geometry1D" type="gml:CurvePropertyType"/>
    </sequence>
   </complexType>
   <complexType name="CellBoundaryGeometryPropertyType">
    <sequence>
     <element ref="core:CellBoundaryGeometry"/>
    </sequence>
   </complexType>
   <element name="CellSpace" substitutionGroup="gml:AbstractFeature" type="core:CellSpaceType"/>
   <complexType name="CellSpaceType">
    <complexContent>
     <extension base="gml:AbstractFeatureType">
      <sequence>
       <element minOccurs="0" name="cellSpaceGeom" type="core:CellSpaceGeometryPropertyType"/>
       <element minOccurs="0" name="externalReference" type="core:ExternalReferencePropertyType"/>
       <element minOccurs="0" name="level" type="string"/>
       <element name="PoI" type="boolean"/>
       <element minOccurs="0" name="cellSpaceName" type="string"/>
       <element minOccurs="0" name="duality" type="core:NodePropertyType">
        <annotation>
         <documentation>Corresponding cell space in the primal space</documentation>
         <appinfo>
          <gml:reversePropertyName>core:duality</gml:reversePropertyName>
         </appinfo>
        </annotation>
       </element>
       <element maxOccurs="unbounded" minOccurs="0" name="boundedBy" type="core:
CellBoundaryPropertyType">
        <annotation>
         <documentation>Cell boundaries that are bounding a cell space</documentation>
        </annotation>
       </element>
      </sequence>
     </extension>
    </complexContent>
   </complexType>
   <complexType name="CellSpacePropertyType">
    <sequence minOccurs="0">
     <element ref="core:CellSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
   </complexType>
   <element name="CellSpaceGeometry" substitutionGroup="gml:AbstractObject" type="core:
CellSpaceGeometryType"/>
   <complexType name="CellSpaceGeometryType">
    <sequence>
```

```
    <element minOccurs="0" name="Geometry3D" type="gml:SolidPropertyType"/>
    <element minOccurs="0" name="Geometry2D" type="gml:SurfacePropertyType"/>
   </sequence>
  </complexType>
  <complexType name="CellSpaceGeometryPropertyType">
   <sequence>
    <element ref="core:CellSpaceGeometry"/>
   </sequence>
  </complexType>
  <element name="DualSpaceLayer" substitutionGroup="gml:AbstractFeature" type="core:
DualSpaceLayerType"/>
  <complexType name="DualSpaceLayerType">
   <complexContent>
    <extension base="gml:AbstractFeatureType">
     <sequence>
      <element name="isLogical" type="boolean"/>
      <element minOccurs="0" name="creationDate" type="dateTime"/>
      <element minOccurs="0" name="terminationDate" type="dateTime"/>
      <element name="isDirected" type="boolean"/>
      <element maxOccurs="unbounded" minOccurs="0" name="edgeMember">
       <annotation>
        <documentation>Edges of a dual space</documentation>
       </annotation>
       <complexType>
        <complexContent>
         <extension base="gml:AbstractFeatureMemberType">
          <sequence minOccurs="0">
           <element ref="core:Edge"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
      <element maxOccurs="unbounded" name="nodeMember">
       <annotation>
        <documentation>Nodes of a dual space</documentation>
       </annotation>
       <complexType>
        <complexContent>
         <extension base="gml:AbstractFeatureMemberType">
          <sequence minOccurs="0">
           <element ref="core:Node"/>
          </sequence>
          <attributeGroup ref="gml:AssociationAttributeGroup"/>
         </extension>
        </complexContent>
       </complexType>
      </element>
     </sequence>
    </extension>
   </complexContent>
  </complexType>
  <complexType name="DualSpaceLayerPropertyType">
   <sequence minOccurs="0">
    <element ref="core:DualSpaceLayer"/>
   </sequence>
   <attributeGroup ref="gml:AssociationAttributeGroup"/>
   <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="Edge" substitutionGroup="gml:AbstractFeature" type="core:EdgeType"/>
  <complexType name="EdgeType">
   <complexContent>
```

```
      <extension base="gml:AbstractFeatureType">
       <sequence>
        <element minOccurs="0" name="geometry" type="gml:CurvePropertyType"/>
        <element name="weight" type="double"/>
        <element minOccurs="0" name="duality" type="core:CellBoundaryPropertyType">
         <annotation>
          <documentation>Corresponding edge in the dual space</documentation>
          <appinfo>
           <gml:reversePropertyName>core:duality</gml:reversePropertyName>
          </appinfo>
         </annotation>
        </element>
        <element maxOccurs="2" minOccurs="2" name="connects" type="core:NodePropertyType">
         <annotation>
          <documentation>The two nodes connected by an edge</documentation>
          <appinfo>
           <gml:reversePropertyName>core:connects</gml:reversePropertyName>
          </appinfo>
         </annotation>
        </element>
       </sequence>
      </extension>
     </complexContent>
    </complexType>
    <complexType name="EdgePropertyType">
     <sequence minOccurs="0">
      <element ref="core:Edge"/>
     </sequence>
     <attributeGroup ref="gml:AssociationAttributeGroup"/>
     <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
    <element name="ExternalObjectReference" substitutionGroup="gml:AbstractObject" type="core:
ExternalObjectReferenceType"/>
    <complexType name="ExternalObjectReferenceType">
     <sequence>
      <element name="name" type="string"/>
      <element name="uri" type="anyURI"/>
     </sequence>
    </complexType>
    <complexType name="ExternalObjectReferencePropertyType">
     <sequence>
      <element ref="core:ExternalObjectReference"/>
     </sequence>
    </complexType>
    <element name="ExternalReference" substitutionGroup="gml:AbstractObject" type="core:
ExternalReferenceType"/>
    <complexType name="ExternalReferenceType">
     <sequence>
      <element name="externalObject" type="core:ExternalObjectReferencePropertyType"/>
      <element name="informationSystem" type="gmd:URL_PropertyType"/>
     </sequence>
    </complexType>
    <complexType name="ExternalReferencePropertyType">
     <sequence>
      <element ref="core:ExternalReference"/>
     </sequence>
    </complexType>
    <element name="IndoorFeatures" substitutionGroup="gml:AbstractFeature" type="core:
IndoorFeaturesType"/>
    <complexType name="IndoorFeaturesType">
     <complexContent>
      <extension base="gml:AbstractFeatureType">
       <sequence>
```

```
        <element maxOccurs="unbounded" name="layers">
         <complexType>
          <complexContent>
           <extension base="gml:AbstractFeatureMemberType">
            <sequence minOccurs="0">
             <element ref="core:ThematicLayer"/>
            </sequence>
            <attributeGroup ref="gml:AssociationAttributeGroup"/>
           </extension>
          </complexContent>
         </complexType>
        </element>
        <element maxOccurs="unbounded" minOccurs="0" name="layerConnections">
         <complexType>
          <complexContent>
           <extension base="gml:AbstractFeatureMemberType">
            <sequence minOccurs="0">
             <element ref="core:InterLayerConnection"/>
            </sequence>
            <attributeGroup ref="gml:AssociationAttributeGroup"/>
           </extension>
          </complexContent>
         </complexType>
        </element>
       </sequence>
      </extension>
     </complexContent>
    </complexType>
    <complexType name="IndoorFeaturesPropertyType">
     <sequence minOccurs="0">
      <element ref="core:IndoorFeatures"/>
     </sequence>
     <attributeGroup ref="gml:AssociationAttributeGroup"/>
     <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
    <element name="InterLayerConnection" substitutionGroup="gml:AbstractFeature" type="core:
InterLayerConnectionType">
     <annotation>
      <documentation>Describes the connection between two thematic layers.</documentation>
     </annotation>
    </element>
    <complexType name="InterLayerConnectionType">
     <complexContent>
      <extension base="gml:AbstractFeatureType">
       <sequence>
        <element name="typeOfTopoExpression" type="core:TopoExpressionValueType">
         <annotation>
          <documentation>Describes the topological relationship between two layers (e.g. overlaps,
contains, etc.).</documentation>
         </annotation>
        </element>
        <element minOccurs="0" name="comment" type="string"/>
        <element maxOccurs="2" minOccurs="2" name="connectedLayers" type="core:
ThematicLayerPropertyType"/>
        <element maxOccurs="2" minOccurs="0" name="connectedCells" type="core:
CellSpacePropertyType">
         <annotation>
          <documentation>Indicates the interconnected primal spaces</documentation>
         </annotation>
        </element>
        <element maxOccurs="2" minOccurs="0" name="connectedNodes" type="core:
NodePropertyType">
         <annotation>
```

```
          <documentation>Indicates the interconnected dual spaces</documentation>
         </annotation>
        </element>
       </sequence>
      </extension>
     </complexContent>
    </complexType>
    <complexType name="InterLayerConnectionPropertyType">
     <sequence minOccurs="0">
      <element ref="core:InterLayerConnection"/>
     </sequence>
     <attributeGroup ref="gml:AssociationAttributeGroup"/>
     <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
    <element name="Node" substitutionGroup="gml:AbstractFeature" type="core:NodeType"/>
    <complexType name="NodeType">
     <complexContent>
      <extension base="gml:AbstractFeatureType">
       <sequence>
        <element minOccurs="0" name="geometry" type="gml:PointPropertyType"/>
        <element name="duality" type="core:CellSpacePropertyType">
         <annotation>
          <documentation>Corresponding node in the dual space</documentation>
          <appinfo>
           <gml:reversePropertyName>core:duality</gml:reversePropertyName>
          </appinfo>
         </annotation>
        </element>
        <element maxOccurs="unbounded" minOccurs="0" name="connects" type="core:
EdgePropertyType">
         <annotation>
          <documentation>Edges connected to the node</documentation>
          <appinfo>
           <gml:reversePropertyName>core:connects</gml:reversePropertyName>
          </appinfo>
         </annotation>
        </element>
       </sequence>
      </extension>
     </complexContent>
    </complexType>
    <complexType name="NodePropertyType">
     <sequence minOccurs="0">
      <element ref="core:Node"/>
     </sequence>
     <attributeGroup ref="gml:AssociationAttributeGroup"/>
     <attributeGroup ref="gml:OwnershipAttributeGroup"/>
    </complexType>
    <element name="PrimalSpaceLayer" substitutionGroup="gml:AbstractFeature" type="core:
PrimalSpaceLayerType"/>
    <complexType name="PrimalSpaceLayerType">
     <complexContent>
      <extension base="gml:AbstractFeatureType">
       <sequence>
        <element minOccurs="0" name="creationDate" type="dateTime"/>
        <element minOccurs="0" name="terminationDate" type="dateTime"/>
        <element maxOccurs="unbounded" name="cellSpaceMember">
         <annotation>
          <documentation>Cell spaces of a primal space</documentation>
         </annotation>
         <complexType>
          <complexContent>
           <extension base="gml:AbstractFeatureMemberType">
```

```
            <sequence minOccurs="0">
              <element ref="core:CellSpace"/>
            </sequence>
            <attributeGroup ref="gml:AssociationAttributeGroup"/>
          </extension>
        </complexContent>
      </complexType>
    </element>
    <element maxOccurs="unbounded" minOccurs="0" name="cellBoundaryMember">
      <annotation>
        <documentation>Cell boundaries of a primal space</documentation>
      </annotation>
      <complexType>
        <complexContent>
          <extension base="gml:AbstractFeatureMemberType">
            <sequence minOccurs="0">
              <element ref="core:CellBoundary"/>
            </sequence>
            <attributeGroup ref="gml:AssociationAttributeGroup"/>
          </extension>
        </complexContent>
      </complexType>
    </element>
  </sequence>
 </extension>
 </complexContent>
</complexType>
<complexType name="PrimalSpaceLayerPropertyType">
 <sequence minOccurs="0">
  <element ref="core:PrimalSpaceLayer"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<element name="ThematicLayer" substitutionGroup="gml:AbstractFeature" type="core:
ThematicLayerType">
 <annotation>
  <documentation>A layer of specific theme aggregating a primal and/or a dual space of a given
environment.</documentation>
 </annotation>
</element>
<complexType name="ThematicLayerType">
 <complexContent>
  <extension base="gml:AbstractFeatureType">
   <sequence>
    <element name="semanticExtension" type="boolean">
     <annotation>
      <documentation>Indicates whether semantic information is associated (true) or not (false) to
the primal space of the thematic layer.</documentation>
     </annotation>
    </element>
    <element name="theme" type="core:ThemeLayerValueType">
     <annotation>
      <documentation>Determines the theme of the layer (e.g., topographic, logical, etc.).</
documentation>
     </annotation>
    </element>
    <element minOccurs="0" name="primalSpace">
     <annotation>
      <documentation>Relates the primal spaces to the thematicLayer</documentation>
     </annotation>
     <complexType>
      <complexContent>
```

```
            <extension base="gml:AbstractFeatureMemberType">
              <sequence minOccurs="0">
                <element ref="core:PrimalSpaceLayer"/>
              </sequence>
              <attributeGroup ref="gml:AssociationAttributeGroup"/>
            </extension>
          </complexContent>
        </complexType>
      </element>
      <element minOccurs="0" name="dualSpace">
        <annotation>
          <documentation>Relates the dual spaces to the thematicLayer.</documentation>
        </annotation>
        <complexType>
          <complexContent>
            <extension base="gml:AbstractFeatureMemberType">
              <sequence minOccurs="0">
                <element ref="core:DualSpaceLayer"/>
              </sequence>
              <attributeGroup ref="gml:AssociationAttributeGroup"/>
            </extension>
          </complexContent>
        </complexType>
      </element>
    </sequence>
  </extension>
 </complexContent>
</complexType>
<complexType name="ThematicLayerPropertyType">
 <sequence minOccurs="0">
  <element ref="core:ThematicLayer"/>
 </sequence>
 <attributeGroup ref="gml:AssociationAttributeGroup"/>
 <attributeGroup ref="gml:OwnershipAttributeGroup"/>
</complexType>
<simpleType name="ThemeLayerValueType">
 <union memberTypes="core:ThemeLayerValueEnumerationType core:ThemeLayerValueOtherType"/>
</simpleType>
<simpleType name="ThemeLayerValueEnumerationType">
 <restriction base="string">
  <enumeration value="Tags"/>
  <enumeration value="Virtual"/>
  <enumeration value="Unknown"/>
  <enumeration value="Physical"/>
 </restriction>
</simpleType>
<simpleType name="ThemeLayerValueOtherType">
 <restriction base="string">
  <pattern value="other: \w{2,}"/>
 </restriction>
</simpleType>
<simpleType name="TopoExpressionValueType">
 <union memberTypes="core:TopoExpressionValueEnumerationType core:
TopoExpressionValueOtherType"/>
</simpleType>
<simpleType name="TopoExpressionValueEnumerationType">
 <restriction base="string">
  <enumeration value="Contains"/>
  <enumeration value="Overlaps"/>
  <enumeration value="Equals"/>
  <enumeration value="Within"/>
  <enumeration value="Crosses"/>
  <enumeration value="Other"/>
```

```
    </restriction>
  </simpleType>
  <simpleType name="TopoExpressionValueOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
</schema>
```

Listing B.1

B.2. IndoorGML Indoor Navigation Module

```
<?xml version="1.0" encoding="UTF-8"?>
<schema
    xmlns="http://www.w3.org/2001/XMLSchema"
 xmlns:navi="http://www.opengis.net/indoorgml/2.0/navi"
 xmlns:core="http://www.opengis.net/indoorgml/2.0/core"
 xmlns:gml="http://www.opengis.net/gml/3.2"
 elementFormDefault="qualified"
 targetNamespace="http://www.opengis.net/indoorgml/2.0/navi"
 version="2.0">
  <annotation>
    <appinfo>
      <gml:gmlProfileSchema>IndoorGML</gml:gmlProfileSchema>
    </appinfo>
  </annotation>
  <import namespace="http://www.isotc211.org/2005/gmd" schemaLocation="http://schemas.opengis.
net/iso/19139/20070417/gmd/gmd.xsd"/>
  <import namespace="http://www.opengis.net/gml/3.2" schemaLocation="http://schemas.opengis.net/
gml/3.2.1/gml.xsd"/>
  <import namespace="http://www.opengis.net/indoorgml/2.0/core" schemaLocation="IndoorGML_core.
xsd"/>
  <!--XML Schema document created by ShapeChange - http://shapechange.net/-->
  <element name="GeneralSpace" substitutionGroup="navi:NavigableSpace" type="navi:
GeneralSpaceType"/>
  <complexType name="GeneralSpaceType">
    <complexContent>
      <extension base="navi:NavigableSpaceType">
        <sequence>
          <element name="function" type="navi:GeneralSpaceFunctionType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="GeneralSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:GeneralSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="GeneralSpaceFunctionType">
    <union memberTypes="navi:GeneralSpaceFunctionEnumerationType navi:
GeneralSpaceFunctionOtherType"/>
  </simpleType>
  <simpleType name="GeneralSpaceFunctionEnumerationType">
```

```
    <restriction base="string">
      <enumeration value="Administration"/>
      <enumeration value="Business, trade"/>
      <enumeration value="Education, training"/>
      <enumeration value="Recreation"/>
      <enumeration value="Laboratory"/>
      <enumeration value="Storage"/>
      <enumeration value="Security"/>
    </restriction>
  </simpleType>
  <simpleType name="GeneralSpaceFunctionOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <simpleType name="LocomotionAccessType">
    <union memberTypes="navi:LocomotionAccessEnumerationType navi:LocomotionAccessOtherType"/
>
  </simpleType>
  <simpleType name="LocomotionAccessEnumerationType">
    <restriction base="string">
      <enumeration value="Walking"/>
      <enumeration value="Flying"/>
      <enumeration value="Rolling"/>
      <enumeration value="Unspecified"/>
    </restriction>
  </simpleType>
  <simpleType name="LocomotionAccessOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <element name="NavigableBoundary" substitutionGroup="core:CellBoundary" type="navi:
NavigableBoundaryType"/>
  <complexType name="NavigableBoundaryType">
    <complexContent>
      <extension base="core:CellBoundaryType">
        <sequence>
          <element name="NavigableBoundaryFunction" type="navi:NavigableBoundaryFunctionType"/>
          <element minOccurs="0" name="boundaryOrientation" type="boolean"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NavigableBoundaryPropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NavigableBoundary"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="NavigableBoundaryFunctionType">
    <union memberTypes="navi:NavigableBoundaryFunctionEnumerationType navi:NavigableBoundaryF
unctionOtherType"/>
  </simpleType>
  <simpleType name="NavigableBoundaryFunctionEnumerationType">
    <restriction base="string">
      <enumeration value="AnchorBoundary"/>
      <enumeration value="ConnectionBoundary"/>
    </restriction>
  </simpleType>
  <simpleType name="NavigableBoundaryFunctionOtherType">
    <restriction base="string">
```

```
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <element name="NavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NavigableSpaceType"/>
  <complexType name="NavigableSpaceType">
    <complexContent>
      <extension base="core:CellSpaceType">
        <sequence>
          <element name="locomotionType" type="navi:LocomotionAccessType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NavigableSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NavigableSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="NonNavigableBoundary" substitutionGroup="core:CellBoundary" type="navi:
NonNavigableBoundaryType"/>
  <complexType name="NonNavigableBoundaryType">
    <complexContent>
      <extension base="core:CellBoundaryType">
        <sequence/>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NonNavigableBoundaryPropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NonNavigableBoundary"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="NonNavigableSpace" substitutionGroup="core:CellSpace" type="navi:
NonNavigableSpaceType"/>
  <complexType name="NonNavigableSpaceType">
    <complexContent>
      <extension base="core:CellSpaceType">
        <sequence/>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="NonNavigableSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:NonNavigableSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="ObjectSpace" substitutionGroup="navi:NonNavigableSpace" type="navi:
ObjectSpaceType"/>
  <complexType name="ObjectSpaceType">
    <complexContent>
      <extension base="navi:NonNavigableSpaceType">
        <sequence>
          <element minOccurs="0" name="description" type="string"/>
          <element minOccurs="0" name="containedFeatures" type="integer"/>
        </sequence>
      </extension>
```

```
    </complexContent>
  </complexType>
  <complexType name="ObjectSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:ObjectSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <element name="Route" substitutionGroup="gml:AbstractFeature" type="navi:RouteType"/>
  <complexType name="RouteType">
    <complexContent>
      <extension base="gml:AbstractFeatureType">
        <sequence>
          <element minOccurs="0" name="creationDate" type="dateTime"/>
          <element maxOccurs="unbounded" minOccurs="2" name="routeNode" type="core:
NodePropertyType"/>
          <element maxOccurs="unbounded" name="routeEdge" type="core:EdgePropertyType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="RoutePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:Route"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="TransferSpaceType">
    <union memberTypes="navi:TransferSpaceEnumerationType navi:TransferSpaceOtherType"/>
  </simpleType>
  <simpleType name="TransferSpaceEnumerationType">
    <restriction base="string">
      <enumeration value="Door"/>
      <enumeration value="Window"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
  <element name="TransferSpace" substitutionGroup="navi:NavigableSpace" type="navi:
TransferSpaceType"/>
  <complexType name="TransferSpaceType">
    <complexContent>
      <extension base="navi:NavigableSpaceType">
        <sequence>
          <element name="function" type="navi:TransferSpaceFunctionType"/>
          <element name="type" type="navi:TransferSpaceType"/>
        </sequence>
      </extension>
    </complexContent>
  </complexType>
  <complexType name="TransferSpacePropertyType">
    <sequence minOccurs="0">
      <element ref="navi:TransferSpace"/>
    </sequence>
    <attributeGroup ref="gml:AssociationAttributeGroup"/>
    <attributeGroup ref="gml:OwnershipAttributeGroup"/>
  </complexType>
  <simpleType name="TransferSpaceFunctionType">
```

```
    <union memberTypes="navi:TransferSpaceFunctionEnumerationType navi:TransferSpaceFunctionOth
erType"/>
  </simpleType>
  <simpleType name="TransferSpaceFunctionEnumerationType">
    <restriction base="string">
      <enumeration value="ConnectionSpace"/>
      <enumeration value="AnchorSpace"/>
    </restriction>
  </simpleType>
  <simpleType name="TransferSpaceFunctionOtherType">
    <restriction base="string">
      <pattern value="other: \w{2,}"/>
    </restriction>
  </simpleType>
</schema>
```

Listing B.2

C

ANNEX C (INFORMATIVE) EXAMPLE OF CODELIST

ANNEX C (INFORMATIVE) EXAMPLE OF CODELIST

For mapping indoor space features to IndoorNavigation module classes, space features have to be classified by functions and usages of space. Because misspellings or different names for the same notion brings the problems in data interoperability, in order to overcome the typo errors, an example of CodeList is provided in this annex to specify the attribute values including space category types and space function types. The CodeLists are defined from OmniClass (Table 13 and Table 14) created and used by the North American architectural, engineering and construction (AEC) industry.

C.1. GeneralSpace

C.1.1. GeneralSpaceFunctionType

Code list derived from OmniClass Table 13:

- T1000 — Corridor
- T1010 — Breezeway
- T1020 — Box Lobby
- T1030 — Elevator Lobby
- T1040 — Landing
- T1050 — Aisle
- T1060 — Ramp
- T1070 — Concourse
- T1080 — Moving walkway
- T1090 — Entry Lobby
- T1100 — Jet way
- T1110 — Elevator Shaft

- T1120 — Stair
- T1130 — Chute
- G1000 — Elevator Machine Room
- G1010 — Fire Command Center
- G1020 — Men’s Restroom
- G1030 — Unisex Restroom
- G1040 — Refrigerant Machinery Room
- G1050 — Incinerator Room
- G1060 — Gas Room
- G1070 — Liquid Use, Dispensing and Mixing Room
- G1080 — Electrical Room
- G1090 — Telecommunications Room
- G1100 — Hazardous Waste Storage
- G1110 — Building Manager Office
- G1120 — Guard Stations
- G1130 — Women’s Restroom
- G1140 — Furnace Room
- G1150 — Fuel Room
- G1160 — Liquid Storage Room
- G1170 — Hydrogen Cutoff Room
- G1180 — Switch Room
- G1190 — Classrooms
- G1200 — Assembly Hall
- G1210 — Physics Teaching Laboratory
- G1220 — Open Class Laboratory
- G1230 — Laboratory Service Space
- G1240 — Computer Lab
- G1250 — Training Support Space

- G1260 — Study Room
- G1270 — Evidence Room
- G1280 — Witness Stand
- G1290 — Robing Area
- G1300 — Council Chambers
- G1310 — Armory
- G1320 — General Performance Spaces
- G1330 — Orchestra Pit
- G1340 — Performance Hall
- G1350 — Audience Space
- G1360 — Audience Seating Space
- G1370 — Projection Booth
- G1380 — Stage Wings
- G1390 — Art Gallery
- G1400 — Sculpture Garden
- G1410 — Recording Studio
- G1420 — Photo Lab
- G1430 — Library
- G1440 — Mediation Chapel
- G1450 — Reflection Space
- G1460 — Chapel
- G1470 — Shrine
- G1480 — Confessional Space
- G1490 — Tabernacle
- G1500 — Choir Loft
- G1510 — Marriage Sanctuary
- G1520 — Mental Health Quiet Room
- G1530 — Bone Densitometry Room

- G1540 — CT Simulator Room
- G1550 — Head Radiographic Room
- G1560 — Mobile Imaging System Alcove
- G1570 — MRI System Component Room
- G1580 — PET/CT Simulator Room
- G1590 — Radiographic Room
- G1600 — Stereotactic Mammography Room
- G1610 — Ultrasound/Optical Coherence Tomography Room
- G1620 — Angiographic Control Room
- G1630 — Angiographic Procedure Control Area
- G1640 — Silver Collection Area
- G1650 — Computer Image Processing Area
- G1660 — CT Control Area
- G1670 — Image Quality Control Room
- G1680 — X-Ray, Plane Film Storage Space
- G1690 — MRI Control Room
- G1700 — MRI Viewing Room
- G1710 — Radiographic Control Room
- G1720 — Tele-Radiology/Tele-Medicine Room
- G1730 — Radiation Diagnostic and Therapy Spaces
- G1740 — Health Physics Laboratory
- G1750 — Linear Accelerator Entrance Maze, Healthcare
- G1760 — Radioactive Waste Storage Room, Healthcare
- G1770 — Nuclear Medicine Scanning Room
- G1780 — Patient Dose/Thyroid Uptake Room
- G1790 — Radiopharmacy
- G1800 — Radiation Therapy, Mold Fabrication Shop
- G1810 — Hearth and Lung Diagnostic and Treatment Spaces

- G1820 — Cardiac Catheter Instrument Room
- G1830 — Cardiac Catheter Control Room
- G1840 — Cardiac Electrophysiology Room
- G1850 — Echocardiograph Room
- G1860 — Extended Pulmonary Function Testing Laboratory
- G1870 — Pacemaker ICD Interrogation Room
- G1880 — Procedure Viewing Area
- G1890 — Pulmonary Function Treadmill Room
- G1900 — Respiratory Therapy Clean-up Room
- G1910 — General Diagnostic Procedure and Treatment Spaces
- G1920 — Endoscopy/Gastroenterology Spaces
- G1930 — Clinical Laboratory Spaces
- G1940 — Pharmacy Spaces
- G1950 — Rehabilitation Spaces
- G1960 — Medical Research and Development Spaces
- G1970 — Chemistry Laboratories
- G1980 — Physical Sciences Laboratories
- G1990 — Earth and Environmental Sciences Laboratories
- G2000 — Psychology Laboratories
- G2010 — Dry Laboratories
- G2020 — Wet Laboratories
- G2030 — Biosciences Laboratories
- G2040 — Astronomy Laboratories
- G2050 — Forensics Laboratories
- G2060 — Bench Laboratories
- … **(Refer to the full standard implementation schema for other classifications)**

C.2. TransferSpace

C.2.1. TransferSpaceCategoryType

- C1000 — Door
- C1010 — Vestibule
- C1020 — Sally port
- A1000 — Vestibule
- A1020 — Gate

C.2.2. TransferSpaceFunctionType

- C1000 — Door
- C1010 — Vestibule
- C1060 — Sally port
- A1000 — Entry Vestibule
- A1010 — Exterior door
- A1020 — Gate
- A1030 — Emergency door

BIBLIOGRAPHY

BIBLIOGRAPHY

[1] Policy SWG, Open Geospatial Consortium (OGC): OGC 08-131r3, *The Specification Model — Standard for Modular specifications*. Open Geospatial Consortium (2009). https://docs.ogc.org/pol/08-131r3/08-131r3/pdf.

[2] Geographic information – Spatial Schema

[3] OGC: CityGML, OGC 08-007r1 2008

[4] OGC: OGC City Geography Markup Language (CityGML) Encoding Standard version 2.0, OGC 12-019, 2012

[5] OGC: OGC KML version 2.2, OGC 07-147r2, 2008

[6] Lefebvre, S., Hornus S.: Automatic cell-and-portal decomposition, Technical Report 4898, INRIA, 2003

[7] Anagnostopoulus, C., Testsos, V., Kikiras, P., Hadjiefthymiades, S. P.: OntoNav: A Semantic Indoor Navigation System, First International Workshop on Managing Context Information in Mobile and Pervasive Environments, Vol. 165, Ayia Napa, Cyprus, 2003

[8] Kulyukin, V., Gharpure, C., Nicholson, J.: RoboCart: Toward Robot-Assisted Navigation of Grocery Stores by the Visually Impaired, Proceedings of the International Conference on Intelligent Robots and Systems, IROS 2005, IEEE/RSJ

[9] Booch, G., Rumbaugh, J., and Jacobson, I.: Unified Modeling Language User Guide, Addison-Wesley, 1997

[10] Liao, L., Fox, D., Hightower, J., Kautz, H., Schulz, D.: Voronoi Tracking: Location Estimation Using Sparse and Noisy Sensor Data, Proc. of the International Conference on Intelligent Robots and Systems, IROS 2003, IEEE/RSJ

[11] Princeton University, 2010

[12] Adachi, Y., Forester, J., Hyvarinen, J., Karstila, K., Liebich, T., Wix, J.: Industry Foundation Classes IFC2x Edition 3, International Alliance for Interoperability, 2003

[13] buildingSmart: Industry Foundation Classes Release 2×3 version 2

[14] Munkres, J. R.: Elements of Algebraic Topology, Addison-Wesley, Menlo Park, CA, 1984

[15] Egenhofer, M., Herring, J.: Categorizing Binary Topological Relations between Regions, Lines and Points in Geographical Databases, NCGIA Technical Report, 1990

[16] Lee, Y.C.: Geographic information systems for urban applications: Problems and solutions, Environment and Planning B: Planning and Design, Vol. 17:463-473, 1990.

[17] Lee, J.: 3D GIS for Geo-coding Human Activity in Micro-scale Urban Environments, M.J. Egenhofer, C. Freksa, and H.J. Miller (Eds.): GIScience 2004, Springer, Berlin, Germany

[18] Lee, J.: A Spatial Access Oriented Implementation of a Topological Data Model for 3D Urban Entities, GeoInformatica 8(3), 235-262, 2004.

[19] Kolbe, T.H., Gröger, G. & Plümer, L.: CityGML – Interoperable Access to 3D City Models, P. van Oosterom, S. Zlatanova & E.M. Fendel (eds), Geo-information for Disaster Management; Proc. of the 1st International Symposium on Geo-information for Disaster Management, Delft, The Netherlands, March 21–23, 2005. Springer.

[20] Kolodziej, K. W., Hjelm, J.: Local Positioning Systems – LBS Applications and Services, Taylor & Francis Group, London, UK, 2006

[21] LaValle, S. M.: Planning Algorithms, Cambridge University Press, USA, 2006

[22] Lorenz, B., Ohlbach, H. J., Stoffel, E.-P., Rosner, M.: NL Navigation Commands from Indoor WLAN fingerprinting position data, Technical Report of REWERSE-Project, Munich, Germany, September 2006

[23] Gold, C. M. and Ledoux, H.: Simultaneous Storage of Primal and Dual Three-Dimensional Subdivisions, Computers, Environment and Urban Systems – Topology and Spatial Databases, Vol. 31, Issue 4, 2007, Elsevier

[24] Lorenz, B., Ohlbach, H. J., Stoffel, E.-P., Rosner, M.: Towards a Semantic Spatial Model for Pedestrian Indoor Navigation, Lecture Notes in Computer Science — Advances in Conceptual Modeling – Foundations and Applications, Volume 4802/2007, Springer, Berlin, Germany

[25] Morris, Sidney A.: Topology without Tears, University of Ballarat, Australia, 2007

[26] Becker, T., Nagel, C., and Kolbe, T. H.: A Multilayered Space-Event Model for Navigation in Indoor Spaces, Lecture Notes in Geoinformation and Cartography — 3D Geo-Information Science, 2008.

[27] Lee, J., Zlatanova, S.: A 3D data model and topological analyses for emergency response in urban areas. Geospatial Information Technology for Emergency Response – Zlatanova & Li (eds), Taylor & Francis Group, London, UK, 2008

[28] Lee, J.: Indoor Spatial Data Model, Proc. ISA Workshop, Seoul, Korea, Aug. 2008

[29] Fritsch, R. and Piccinini, R. A.: Cellular Structures in Topology, Cambridge University Press, Cambridge, England, 2008

[30] Becker, T., Nagel, C., and Kolbe, T. H.: Supporting Contexts for Indoor Navigation using a Multilayered Space Model, Tenth International Conference on Mobile Data Management, Taipei, Taiwan, 2009

[31] Nagel, C., Becker B., Kaden, R., Li, K-J., Lee, J., and Kolbe, T. H.: Requirements and Space-Event Modeling for Indoor Navigation, OGC 10-191r1, 2010